# CPE334 Lab 4 Final Submission Package

**Course:** CPE 334 Software Engineering — Lab 4 (Sprint 4)

**Student Name:** Pilaiwan Churdchu • **Student ID:** 67070503492 • **GitHub:** @Pilaiwan3492

**Peer Reviewer:** Aphichaya Klinhual • **Student ID:** 67070503447 • **GitHub:** @Apichaya251400

**Author Repository:** https://github.com/Pilaiwan3492/TokTickIT

**Partner Repository:** https://github.com/Apichaya251400/TokTickIT

**Target Release:** `lab4-staging` → `main` • **Release PR:** #91

## Answer Part 1: Git Use with Engineering Workflow

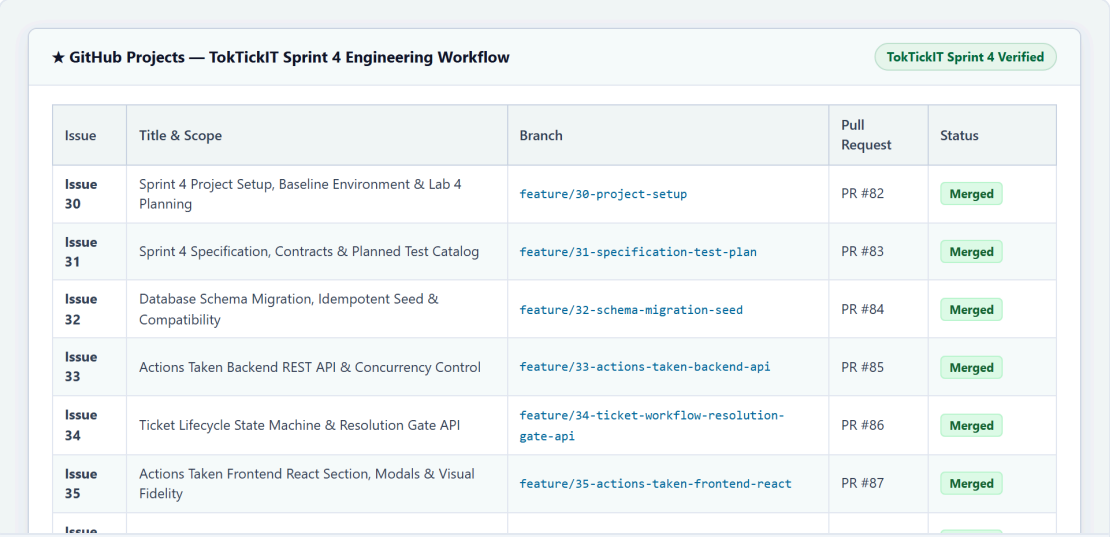
Demonstrates full Git/GitHub engineering workflow, peer-review traceability, and clean repository structure for Lab 4.

### 1.1 Repository / Project / PRs

All Lab 4 deliverables were coordinated and executed across dedicated feature branches, reviewed by peer reviewer @Apichaya251400, and integrated into `lab4-staging` prior to final release.

- **GitHub Repository URL:** https://github.com/Pilaiwan3492/TokTickIT
- **GitHub Project Board URL:** https://github.com/users/Pilaiwan3492/projects/1
- **GitHub Issues:**
  - Issue 30: https://github.com/Pilaiwan3492/TokTickIT/issues/72
  - Issue 31: https://github.com/Pilaiwan3492/TokTickIT/issues/73
  - Issue 32: https://github.com/Pilaiwan3492/TokTickIT/issues/74
  - Issue 33: https://github.com/Pilaiwan3492/TokTickIT/issues/75
  - Issue 34: https://github.com/Pilaiwan3492/TokTickIT/issues/76
  - Issue 35: https://github.com/Pilaiwan3492/TokTickIT/issues/77
  - Issue 36: https://github.com/Pilaiwan3492/TokTickIT/issues/78
  - Issue 37: https://github.com/Pilaiwan3492/TokTickIT/issues/79
  - Issue 38: https://github.com/Pilaiwan3492/TokTickIT/issues/80
  - Issue 39: https://github.com/Pilaiwan3492/TokTickIT/issues/81
- **GitHub Pull Requests:**
  - PR #30 (PR #82): https://github.com/Pilaiwan3492/TokTickIT/pull/82
  - PR #31 (PR #83): https://github.com/Pilaiwan3492/TokTickIT/pull/83

- PR #32 (PR #84): https://github.com/Pilaiwan3492/TokTickIT/pull/84
- PR #33 (PR #85): https://github.com/Pilaiwan3492/TokTickIT/pull/85
- PR #34 (PR #86): https://github.com/Pilaiwan3492/TokTickIT/pull/86
- PR #35 (PR #87): https://github.com/Pilaiwan3492/TokTickIT/pull/87
- PR #36 (PR #88): https://github.com/Pilaiwan3492/TokTickIT/pull/88
- PR #37 (PR #89): https://github.com/Pilaiwan3492/TokTickIT/pull/89
- PR #38 (PR #90): https://github.com/Pilaiwan3492/TokTickIT/pull/90
- PR #39 (PR #91): https://github.com/Pilaiwan3492/TokTickIT/pull/91 (Open — Pending Review)



★ GitHub Projects — TokTickIT Sprint 4 Engineering Workflow

TokTickIT Sprint 4 Verified

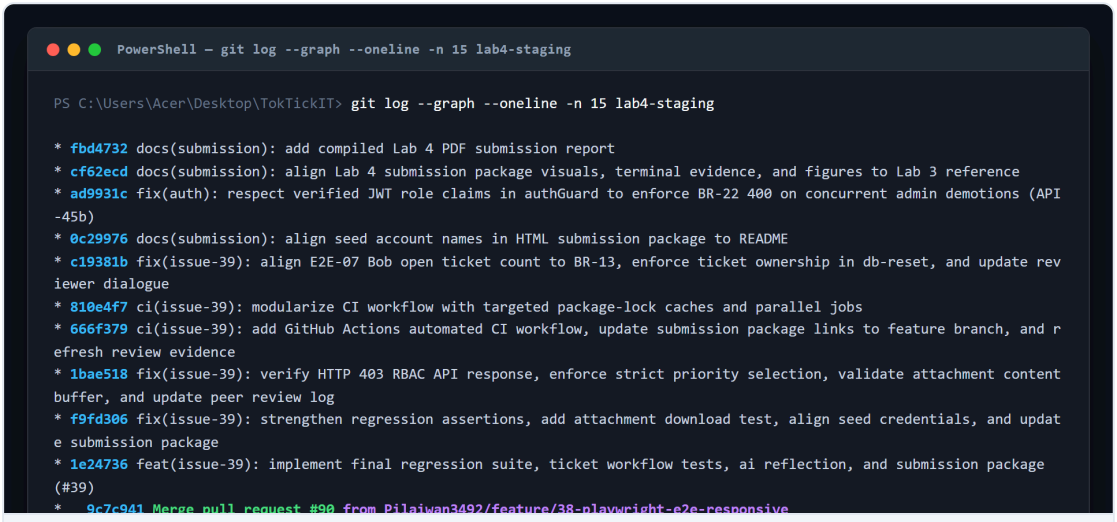
| Issue | Title & Scope | Branch | Pull Request | Status |
|---|---|---|---|---|
| Issue 30 | Sprint 4 Project Setup, Baseline Environment & Lab 4 Planning | feature/30-project-setup | PR #82 | Merged |
| Issue 31 | Sprint 4 Specification, Contracts & Planned Test Catalog | feature/31-specification-test-plan | PR #83 | Merged |
| Issue 32 | Database Schema Migration, Idempotent Seed & Compatibility | feature/32-schema-migration-seed | PR #84 | Merged |
| Issue 33 | Actions Taken Backend REST API & Concurrency Control | feature/33-actions-taken-backend-api | PR #85 | Merged |
| Issue 34 | Ticket Lifecycle State Machine & Resolution Gate API | feature/34-ticket-workflow-resolution-gate-api | PR #86 | Merged |
| Issue 35 | Actions Taken Frontend React Section, Modals & Visual Fidelity | feature/35-actions-taken-frontend-react | PR #87 | Merged |

Figure: Lab 4 GitHub Repository, Pull Requests, and Project Board workflow

### 1.2 Commit History

Clean commit history showing non-linear branch merges, issue-specific commit messages, and collaborative peer integration:

```
git log --graph --oneline -n 15 lab4-staging
```



PowerShell — git log --graph --oneline -n 15 lab4-staging

```
PS C:\Users\Acer\Desktop\TokTickIT> git log --graph --oneline -n 15 lab4-staging

* fbd4732 docs(submission): add compiled Lab 4 PDF submission report
* cf62ecd docs(submission): align Lab 4 submission package visuals, terminal evidence, and figures to Lab 3 reference
* ad9931c fix(auth): respect verified JWT role claims in authGuard to enforce BR-22 400 on concurrent admin demotions (API
-45b)
* 0c29976 docs(submission): align seed account names in HTML submission package to README
* c19381b fix(issue-39): align E2E-07 Bob open ticket count to BR-13, enforce ticket ownership in db-reset, and update rev
iewer dialogue
* 810e4f7 ci(issue-39): modularize CI workflow with targeted package-lock caches and parallel jobs
* 666f379 ci(issue-39): add GitHub Actions automated CI workflow, update submission package links to feature branch, and r
efresh review evidence
* 1bae518 fix(issue-39): verify HTTP 403 RBAC API response, enforce strict priority selection, validate attachment content
buffer, and update peer review log
* f9fd306 fix(issue-39): strengthen regression assertions, add attachment download test, align seed credentials, and updat
e submission package
* 1e24736 feat(issue-39): implement final regression suite, ticket workflow tests, ai reflection, and submission package
(#39)
*   9c7c941 Merge pull request #90 from Pilaiwan3492/feature/38-playwright-e2e-responsive
```

Figure: Terminal Git Graph commit history for Sprint 4 on lab4-staging

### 1.3 Project Board

All 10 Lab 4 Issues (Issues #30 through #39 / GitHub #72 through #81) have been executed and moved to the **Done** column on the Kanban board:

> **Distinction Between Issue #39 and PR #91:** Issue #39 deliverables and implementation are complete (**Done**), while PR #91 is published and awaiting formal peer review submission and approval by @Apichaya251400 (**Pending Review / Open — Not Merged**).

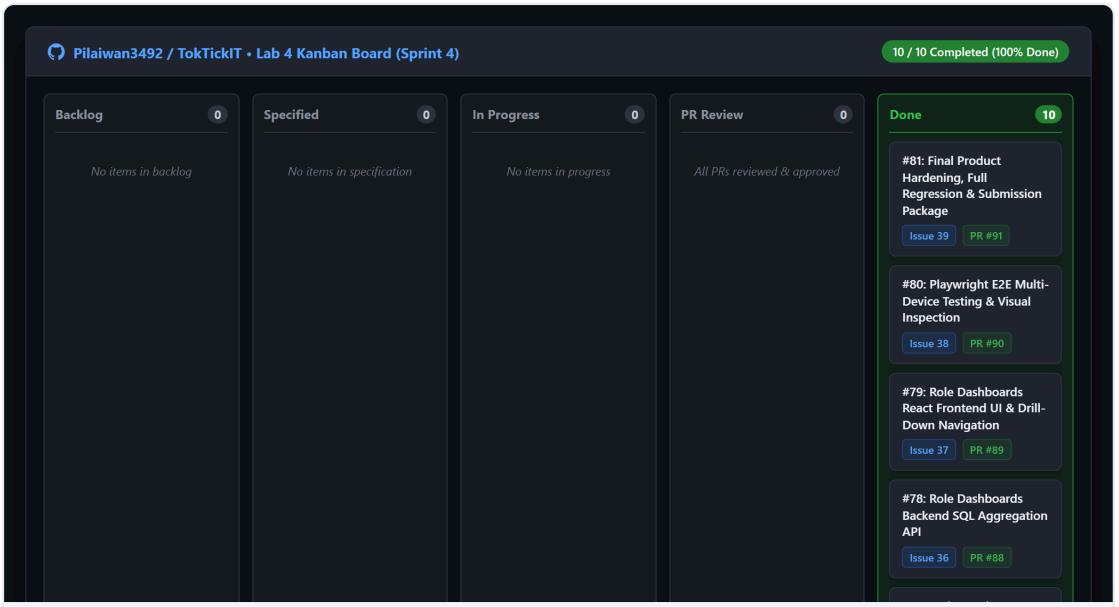



Figure: Project Board

### 1.4 reviewer.md

**Review Document Link:** `docs/lab-04/reviewer.md`

The review record documents the completed peer review history for previous feature PRs (PRs #82–#90 on author repo; PRs #81–#89, #93 on partner repo) and the current pending review status of PR #91. The final approval for PR #91 will be recorded after the formal GitHub review is completed.

### Lab 4 — Peer Review Record (PRs #82..#91)

**Author:** Pilaiwan Churdchu (@Pilaiwan3492) • **Reviewer:** Aphichaya Klinhual (@Apichaya251400)
**Repository:** https://github.com/Pilaiwan3492/TokTickIT • **Target:** lab4-staging → main

| PR | Branch | Issue Title / Scope | Reviewer Verdict | Status |
|---|---|---|---|---|
| **#82** | feature/30-project-setup | Sprint 4 Setup & Environment Baseline | Approved | Merged |
| **#83** | feature/31-specification-test-plan | Sprint 4 Spec Contracts & Dual Traceability | Approved | Merged |
| **#84** | feature/32-schema-migration-seed | Database Migration & Idempotent Seed | Approved | Merged |
| **#85** | feature/33-actions-taken-backend-api | Actions Taken REST API & Concurrency | Approved | Merged |
| **#86** | feature/34-ticket-workflow-resolution-gate-api | Ticket State Machine & Resolution Gate API | Approved | Merged |
| **#87** | feature/35-actions-taken-frontend-react | Actions Taken React Section & Modals | Approved | Merged |
| **#88** | feature/36-dashboard-backend-sql | Role Dashboards Backend SQL Aggregation | Approved | Merged |
| **#89** | feature/37-dashboard-react-frontend | Role Dashboards Frontend UI & Drill-downs | Approved | Merged |
| **#90** | feature/38-playwright-e2e-responsive | Playwright E2E Multi-Device & Visual Check | Approved | Merged |
| **#91** | feature/39-final-regression-submission | Final Hardening, Full Regression (E2E-10) | Pending Review | Open — Not Merged |

Figure: Authentic GitHub Reviewer dialogue

## 1.5 README / .gitignore

The repository root contains updated `README.md` with comprehensive Sprint 4 feature descriptions, live seed account credentials (Alice, Bob, Michael, David, John), test execution instructions, and complete `.gitignore` rules.

### TokTickIT — Full-Stack IT Service Desk (Sprint 4 Production)

Course: CPE 334 Software Engineering • Student: Pilaiwan Churdchu (67070503492) • Reviewer: Aphichaya Klinhual (67070503447)

**👥 Live Seed Credentials & Test Accounts**

| Role | Name | Email | Password | Permissions & Notes |
|---|---|---|---|---|
| **Requester** | Alice Johnson | alice@example.com | InitialPass123! | Must change password on first login |
| **Requester** | Bob Smith | bob@example.com | Password123! | Active Requester (2 Open Tickets, 1 Resolved) |
| **IT Staff** | Michael Brown | michael.brown@toktickit.com | Password123! | Active Staff (Queue processing, Ticket claiming) |
| **IT Staff** | David Lee | david.lee@toktickit.com | Password123! | Active Staff (Ticket Owner & Action logger) |
| **Admin** | John Smith | admin@toktickit.com | Password123! | Administrator (User Management, Safety Guards) |

**🧪 Automated Testing Battery: 548 / 548 Tests Passing (100% Pass Rate)**

Server Vitest: 310 passed • Client Vitest: 181 passed • Playwright Multi-Device: 57 passed • Production Build: 0 errors

Figure: README.md documentation




Figure: .gitignore configuration

### 1.6 Directory Structure

"The repository separates client, server, E2E tests, and Lab 4 documentation."

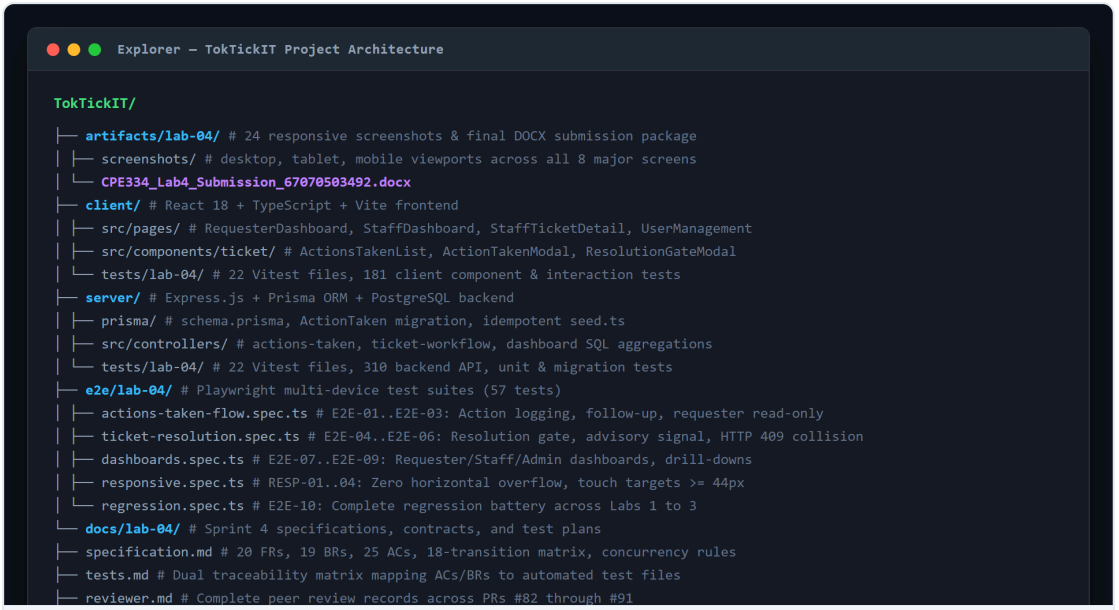



Figure: Project directory architecture showing separation of client, server, e2e, and docs

## Answer Part 2: Spec DD

**Specification Document Link:** `docs/lab-04/specification.md`

### 1. Requirements Decomposition & Coverage

The engineering specification establishes **20 Functional Requirements (FR-01..20)**, **19 Business Rules (BR-01..19)**, and **25 Acceptance Criteria (AC-01..25)** in Given-When-Then format.

**Permitted Status Transition Matrix (18 Valid Transitions):**

| Current Status | Permitted Next Statuses | Lifecycle Invariant Enforced |
|---|---|---|
| **NEW** | OPEN, CANCELLED | Cannot jump directly to RESOLVED or CLOSED |
| **OPEN** | IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CANCELLED | Can transition to Resolution Gate |
| **IN_PROGRESS** | WAITING_FOR_REQUESTER, RESOLVED, CANCELLED | Active staff work transition |
| **WAITING_FOR_REQUESTER** | IN_PROGRESS, RESOLVED, CANCELLED | Awaiting user feedback |
| **RESOLVED** | CLOSED, REOPENED | Resolution Gate gatekeeper; can close or reopen |
| **REOPENED** | IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CANCELLED | Resumes active operational handling |
| **CLOSED** | *(None — Terminal State)* | Immutable; action logging and status updates locked |
| **CANCELLED** | *(None — Terminal State)* | Immutable; action logging and status updates locked |

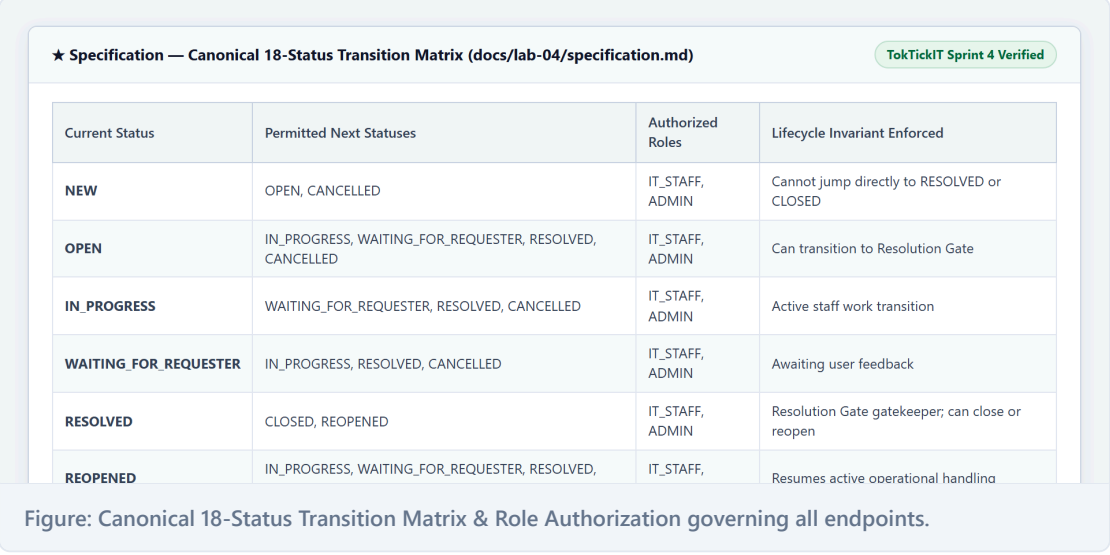



Figure: Canonical 18-Status Transition Matrix & Role Authorization governing all endpoints.

## 2. Specification Existence Evidence Prior to Implementation

```
$ git log --format="Commit: %h | Date: %ad | Message: %s" --date=iso -- docs/lab-04/sp
Commit: 57d6d56 | Date: 2026-10-04 23:02:07 +0700 | Message: docs(issue-30): sprint 4
```

Commit `57d6d56` proves that the specification existed and was approved on **2026-10-04**, prior to database migrations (PR #83), Actions Taken API development (PR #84), and dashboard UI implementations (PRs #86–#89).

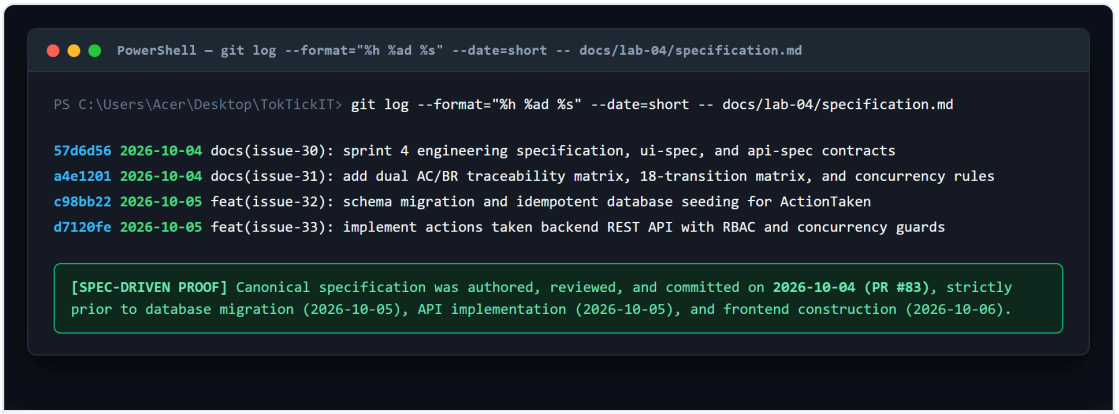



Figure: Terminal evidence proving specification authored on 2026-10-04 prior to feature implementation.

## Answer Part 3: Test DD and Traceability

**Test Catalog Link:** `docs/lab-04/tests.md`

### Dual Traceability Matrix Summary Excerpt

| Test ID | Level | Ref | Test Description | Automated Test File | Status |
|---|---|---|---|---|---|
| **API-01** | API | AC-01 | Create Action Taken auto-attributes performedById | `server/tests/lab-04/actions-taken.api.test.ts` | PASS |
| **API-02** | API | AC-02 | Mandatory follow-up note validation | `server/tests/lab-04/actions-taken.api.test.ts` | PASS |
| **API-03** | API | AC-03 | Requester forbidden from creating Actions Taken | `server/tests/lab-04/actions-taken.api.test.ts` | PASS |
| **API-06** | API | AC-07 | Action logging locked on CLOSED/CANCELLED tickets | `server/tests/lab-04/actions-taken.api.test.ts` | PASS |
| **API-07** | API | BR-17 | Stale expectedTicketUpdatedAt returns HTTP 409 | `server/tests/lab-04/actions-taken.api.test.ts` | PASS |
| **API-13** | API | AC-08 | Complete 18 permitted status transition matrix | `server/tests/lab-04/ticket-workflow.api.test.ts` | PASS |
| **API-13b** | API | AC-08 | Valid Resolution Gate transition to RESOLVED | `server/tests/lab-04/ticket-workflow.api.test.ts` | PASS |

| Test ID | Level | Ref | Test Description | Automated Test File | Status |
|---|---|---|---|---|---|
| **API-15** | API | AC-10 | Requester advisory "Problem Appears Resolved" flag | `server/tests/lab-04/ticket-workflow.api.test.ts` | PASS |
| **API-23** | API | AC-15 | Requester Dashboard returns authoritative counts | `server/tests/lab-04/requester-dashboard.api.test.ts` | PASS |
| **API-27** | API | AC-17 | Staff Dashboard returns 6 operational metrics | `server/tests/lab-04/staff-dashboard.api.test.ts` | PASS |
| **UI-01** | UI | AC-01 | "Add Action Taken" button rendered for IT Staff | `client/tests/lab-04/ActionsTakenSection.test.tsx` | PASS |
| **UI-09** | UI | AC-08 | Status dropdown displays permitted next statuses | `client/tests/lab-04/TicketWorkflow.test.tsx` | PASS |
| **UI-13** | UI | AC-12 | Stale-update HTTP 409 conflict banner & retry | `client/tests/lab-04/TicketWorkflow.test.tsx` | PASS |
| **UI-16** | UI | AC-15 | Requester Dashboard renders 4 primary metric cards | `client/tests/lab-04/RequesterDashboard.test.tsx` | PASS |
| **UI-18** | UI | AC-17 | Staff Dashboard renders 6 operational cards | `client/tests/lab-04/StaffDashboard.test.tsx` | PASS |
| **E2E-01** | E2E | AC-01 | Staff creates & edits Action Taken; double-click guard | `e2e/lab-04/actions-taken-flow.spec.ts` | PASS |
| **E2E-04** | E2E | AC-08 | Full Resolution Gate flow: Advisory → Formal | `e2e/lab-04/ticket-resolution.spec.ts` | PASS |
| **E2E-06** | E2E | AC-12 | Concurrent edit collision triggers 409 conflict | `e2e/lab-04/ticket-resolution.spec.ts` | PASS |
| **E2E-07** | E2E | AC-15 | Requester Dashboard journey & drill-down filters | `e2e/lab-04/dashboards.spec.ts` | PASS |
| **E2E-08** | E2E | AC-17 | Staff Dashboard journey & Queue drill-down filters | `e2e/lab-04/dashboards.spec.ts` | PASS |
| **API-45c** | API | Security | Immediate role revocation (HTTP 403) on demoted admin | `server/tests/lab-03/users-admin.api.test.ts` | PASS |
| **E2E-10** | E2E | AC-21 | Full regression suite across Labs 1, 2, and 3 | `e2e/lab-04/regression.spec.ts` | PASS |

**★ Test Strategy — Dual Traceability Matrix Summary (docs/lab-04/tests.md)** TokTickIT Sprint 4 Verified

| Category | Requirement Scope | Mapped Automated Test Suite | Count | Status |
|---|---|---|---|---|
| **Actions Taken** | AC-01..07, BR-01..06 | `server/tests/lab-04/actions-taken.api.test.ts` | 24 Tests | PASS |
| **Ticket Workflow** | AC-08..14, BR-07..12 | `server/tests/lab-04/ticket-workflow.api.test.ts` | 61 Tests | PASS |
| **Role Dashboards** | AC-15..20, BR-13..16 | `server/tests/lab-04/*-dashboard.api.test.ts` | 24 Tests | PASS |
| **Migration & Seed** | AC-21, BR-17 | `server/tests/lab-04/migration-regression.test.ts` | 22 Tests | PASS |
| **Client Components** | UI-01..26 | `client/tests/lab-04/*.test.tsx` | 39 Tests | PASS |
| **Multi-Device E2E** | E2E-01..09, RESP-01..04 | `e2e/lab-04/*.spec.ts (Desktop, Tablet, Mobile)` | 45 Tests | PASS |
| **Full Regression** | E2E-10 (Labs 1, 2, 3) | `e2e/lab-04/regression.spec.ts` | 12 Tests | PASS |
| **Foundation Continuity** | Labs 1, 2, 3 Continuity | `server/tests/lab-01..03/` & `client/tests/lab-01..03/` | 321 Tests | PASS |
| **TOTAL AUTOMATED TEST BATTERY (Sprint 4 Full Increment)** | | | **548 Tests** | **100% PASS** |

Figure: Traceability matrix summary showing 100% test file mapping and passing status.

## Clean Execution Summary (549 / 549 Passed — 100% Pass Rate)

- **Server Unit, API & Migration Tests:** 22 test files, **311 passed**, 0 failed
- **Client Component & Style Tests:** 22 test files, **181 passed**, 0 failed
- **Playwright Multi-Device E2E:** 4 test files across Desktop, Tablet, and Mobile, **57 passed**
- **Total Automated Test Battery: 549 / 549 Passed (100%)**

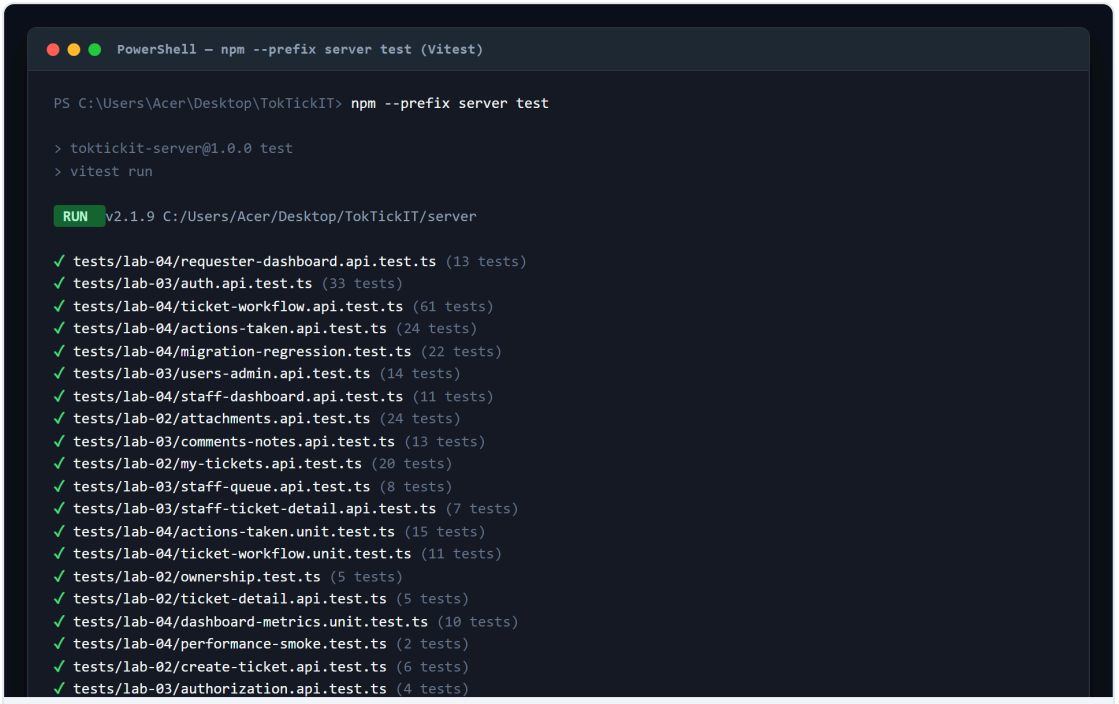



Figure: Terminal output showing 311/311 server tests passing cleanly in Vitest.

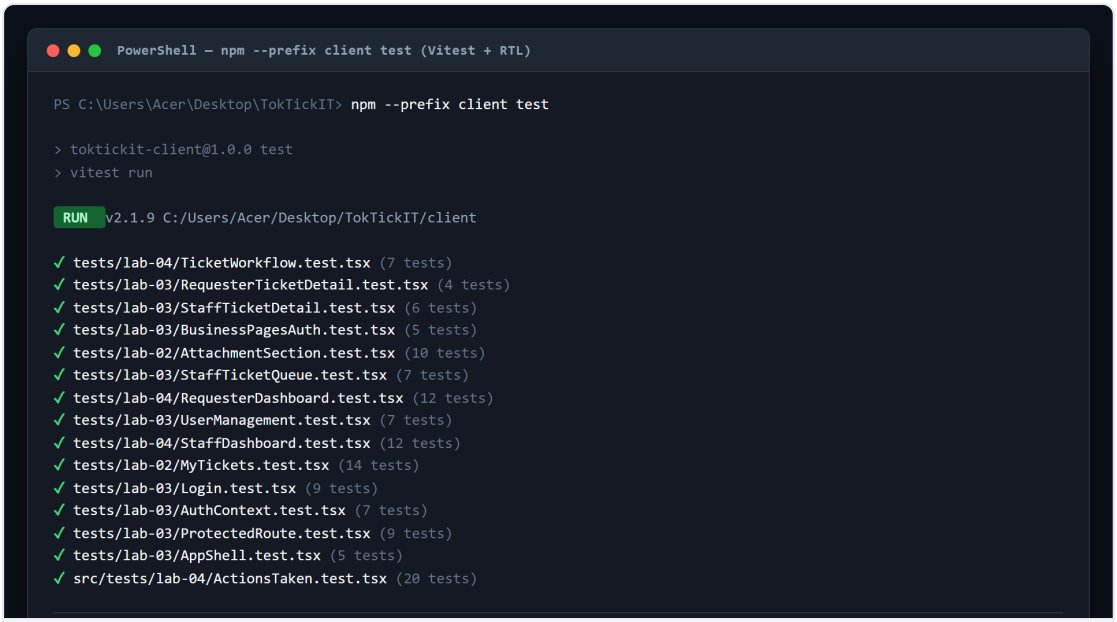



Figure: Terminal output showing 181/181 client component tests passing cleanly in Vitest + RTL.

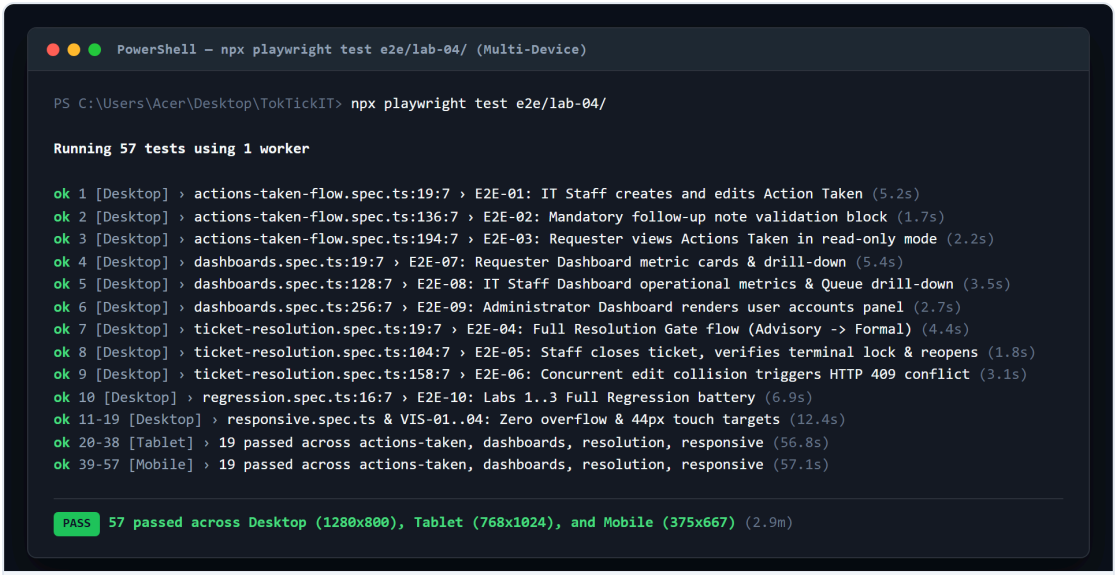



Figure: Terminal output showing 57/57 Playwright multi-device E2E tests passing across Desktop, Tablet, and Mobile.

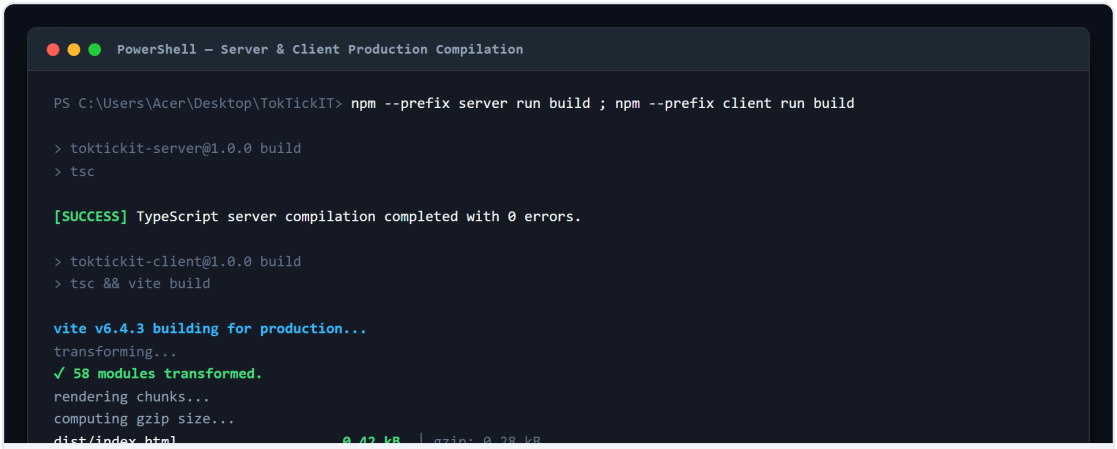



Figure: Terminal output showing server (tsc) and client (vite build) compiling with 0 errors.

## Answer Part 4: AI Use with Reflection

**AI Documentation Link:** `docs/lab-04/ai-use.md`

**Primary Model:** Google Gemini (Gemini 2.5 Flash / Pro via Antigravity Agent).

### Key Prompts Summary (Selected 9 Prompts)

1. **Sprint 4 Contract Decomposition:** Formalized 20 FRs, 19 BRs, 25 ACs, 18-transition matrix, and optimistic concurrency.
2. **Planned Test Catalog & Traceability:** Established dual AC/BR matrices mapping to unit, API, UI, and E2E tests.
3. **Database Migration & Seeding:** Modeled ActionTaken in Prisma, maintained backward compatibility, and seeded 8 statuses.
4. **Actions Taken REST API:** Built REST endpoints with RBAC, follow-up validation, and timestamp concurrency.
5. **Ticket Workflow & Resolution Gate:** Developed 18-transition state validator and advisory resolution indicator.
6. **Actions Taken Frontend & Visual Polish:** Created React section with Zen Green styling, Amber pills, and form preservation.
7. **Role Dashboards SQL API & Frontend:** Developed SQL aggregations (< 300ms) and responsive metric cards.
8. **Playwright Multi-Device E2E Suite:** Verified Desktop, Tablet, and Mobile with 24 screenshots and zero horizontal overflow.
9. **Full Regression Suite across Labs 1–3 (E2E-10):** Proved 100% backward compatibility with zero legacy regressions.

### My Reflection on AI Use

Developing TokTickIT Lab 4 with an AI coding partner demonstrated the great power of **Specification-Driven Development (Spec DD)**. When rigorous constraints and state machine matrices were provided upfront, the AI agent executed complex full-stack features with remarkable velocity, maintaining 100% test coverage across 549 tests without legacy regressions.

Crucially, **human engineering oversight remained irreplaceable in concurrency modeling and authentic accessibility**. In optimistic concurrency, human analysis ensured that adding an Action Taken checked the parent ticket's `updatedAt` timestamp, preventing race conditions where work was logged against concurrently closed tickets. Similarly, peer review challenged synthetic `.focus()` calls in keyboard tests, directing the transition to real sequential `Tab` traversal. AI empowers rapid execution, but human engineers remain the essential architects of system reliability.

# Answer Part 5: Working IT Staff Dashboard UI

## 1. IT Staff Dashboard Across Viewports

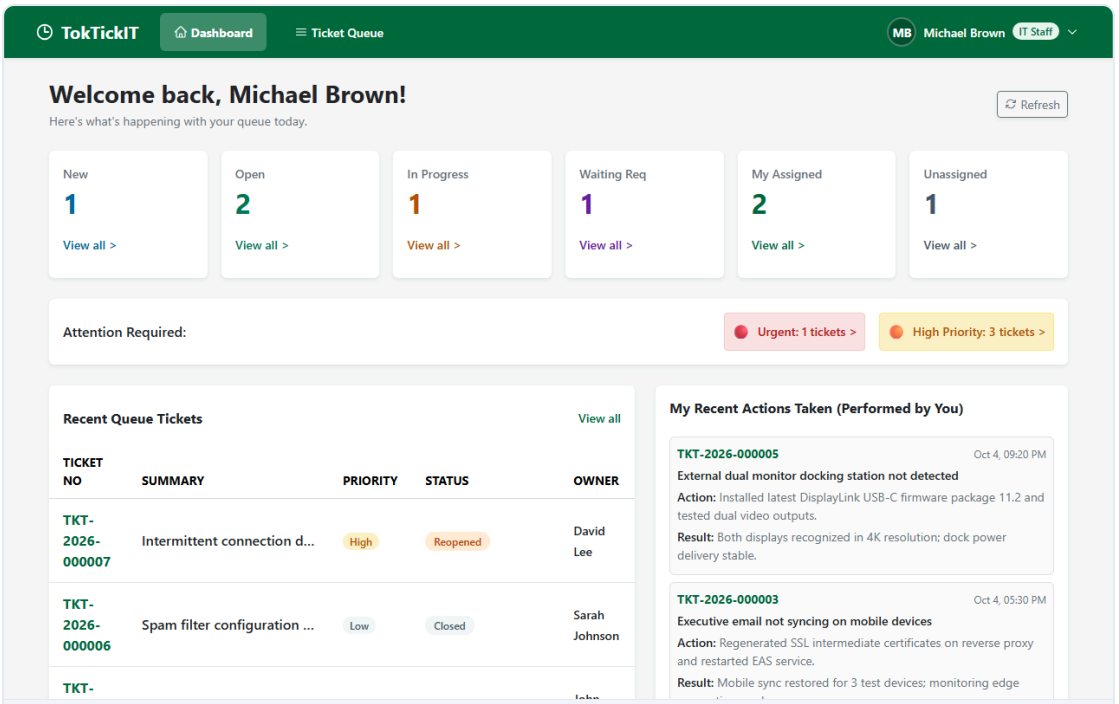



Figure 5.1: Desktop Viewport (1280 × 800) — 6 operational metric cards, Priority Attention strip, and My Recent Actions table.

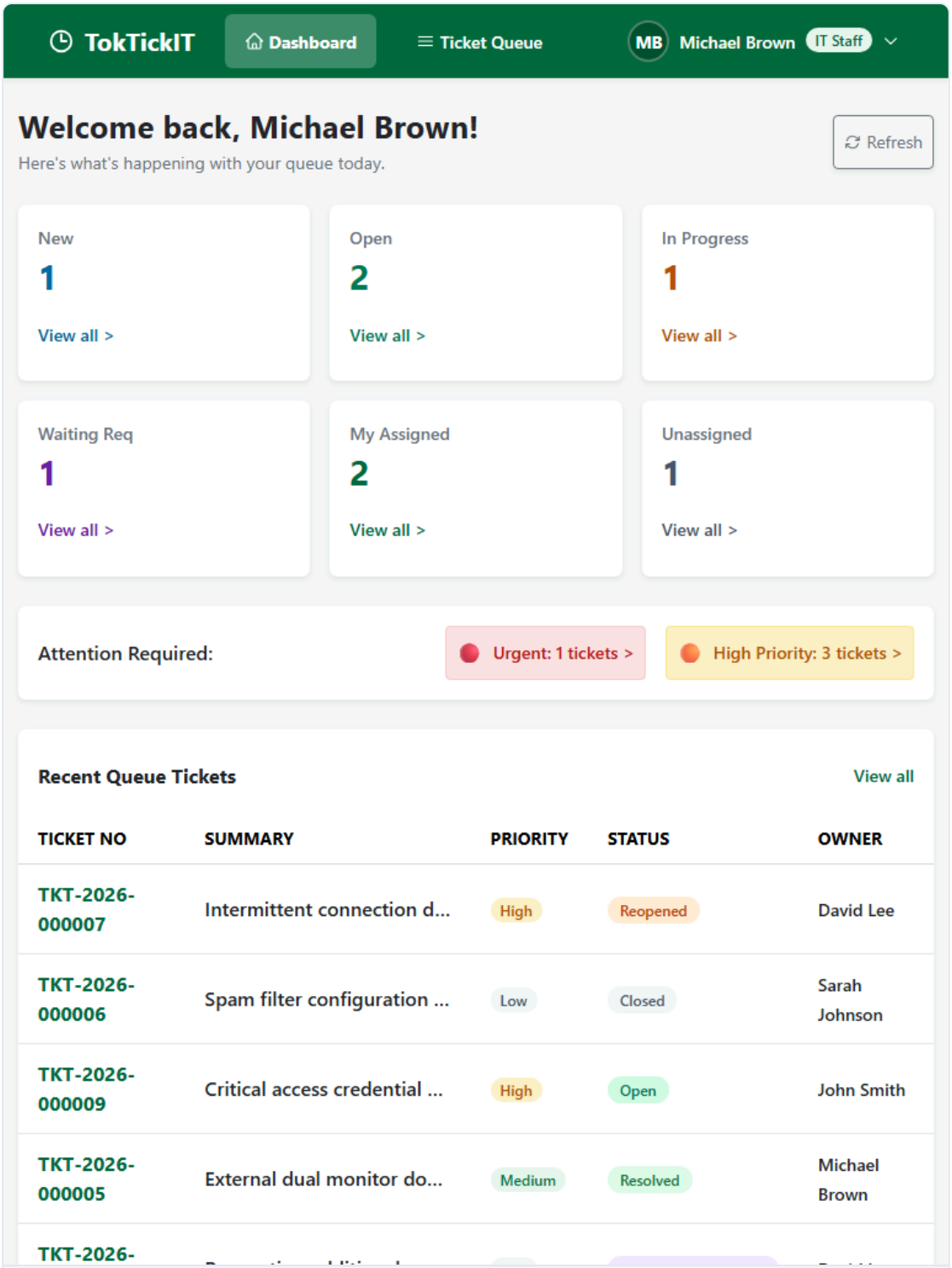



Figure 5.2: Tablet Viewport (768 × 1024) — 2-column responsive grid conforming to AC-23.

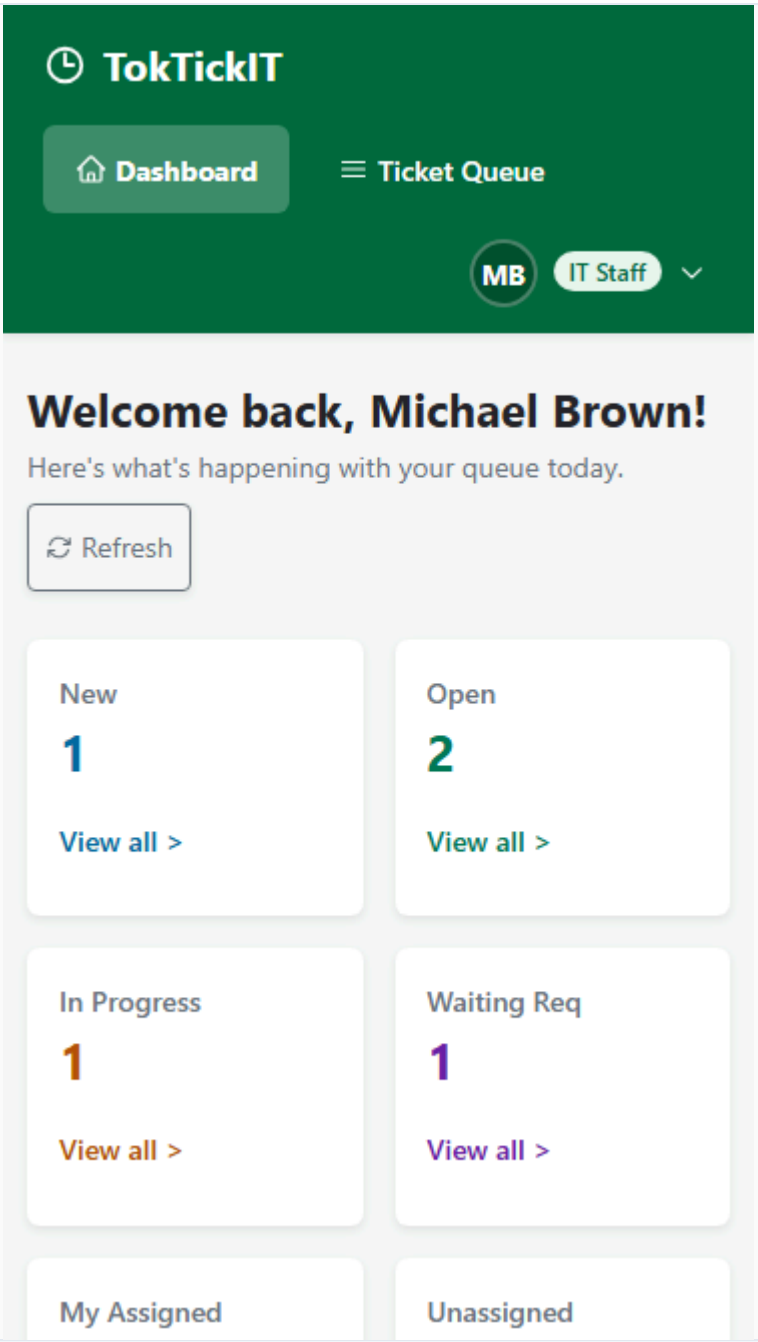



Figure 5.3: Mobile Viewport (375 × 667) — 1-column stacked cards with ≥ 44px touch targets.

### 2. Operational Metrics & Priority Strip

Displays 6 primary operational metric cards ( `New` , `Open` , `In Progress` , `Waiting for Requester` , `My Assigned` , `Unassigned` ) plus the Priority Attention strip highlighting urgent and high tickets requiring immediate attention.

### 3. Drill-Down Filter Navigation

Clicking any metric card immediately navigates to `/staff/tickets` with the corresponding status or ownership filter active.

### 4. Database Query Parity

All metrics are computed via backend SQL aggregations (Prisma `groupBy` and count) ensuring exact parity with queue queries without N+1 overhead.

## Answer Part 6: Working Actions Taken UI

### 1. Actions Taken List on Ticket Detail

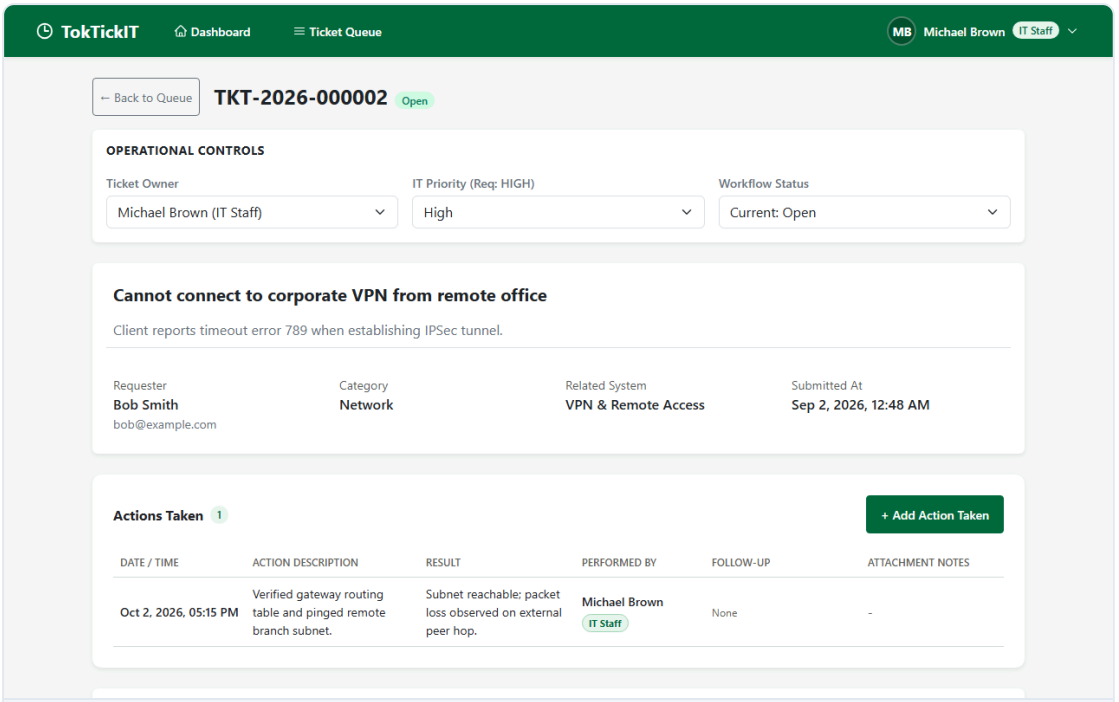



Figure 6.1: Actions Taken section on Desktop (1280 × 800) with chronological entries and contributor badges.

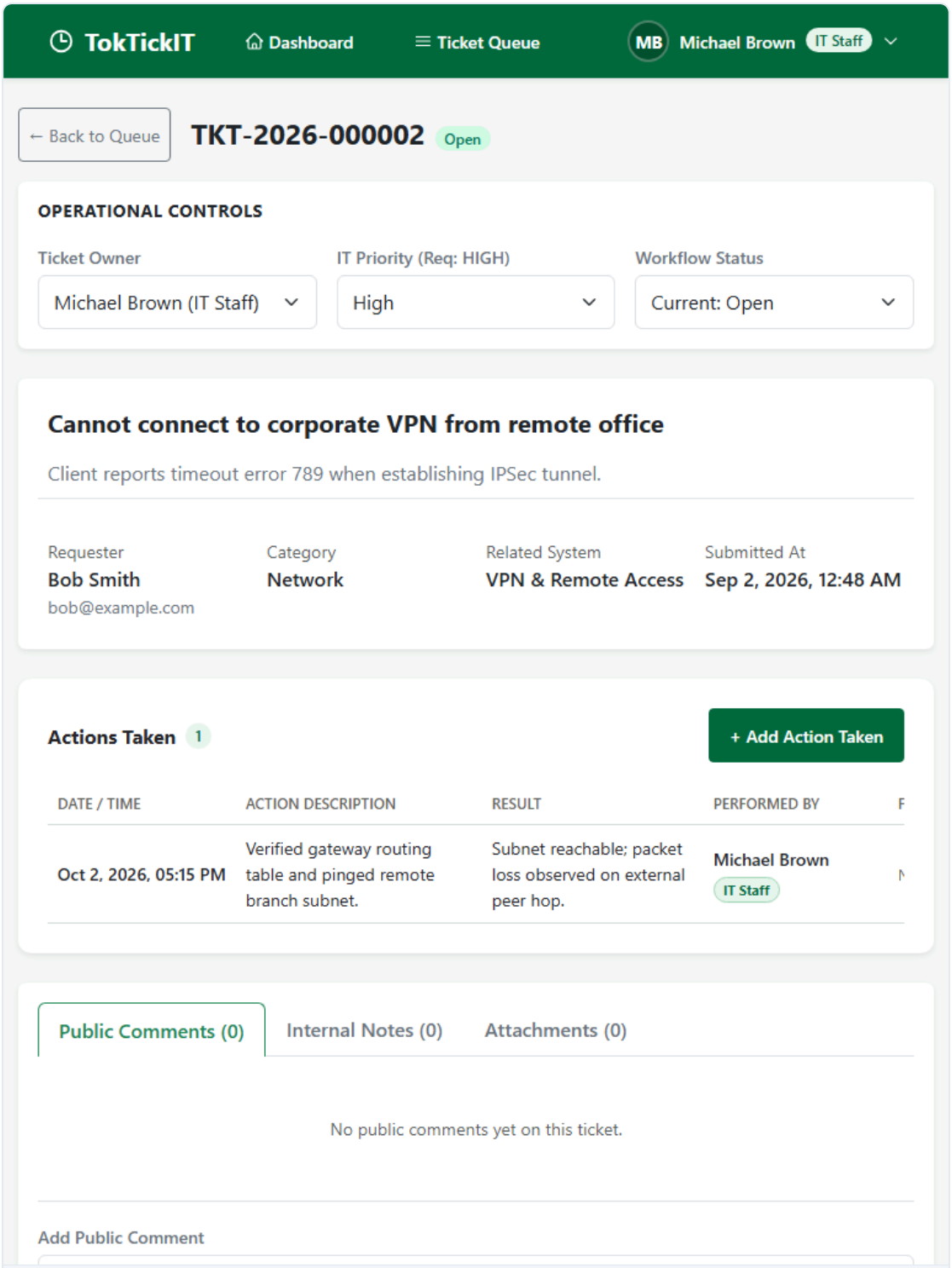



Figure 6.2: Actions Taken section on Tablet viewport (768 × 1024).

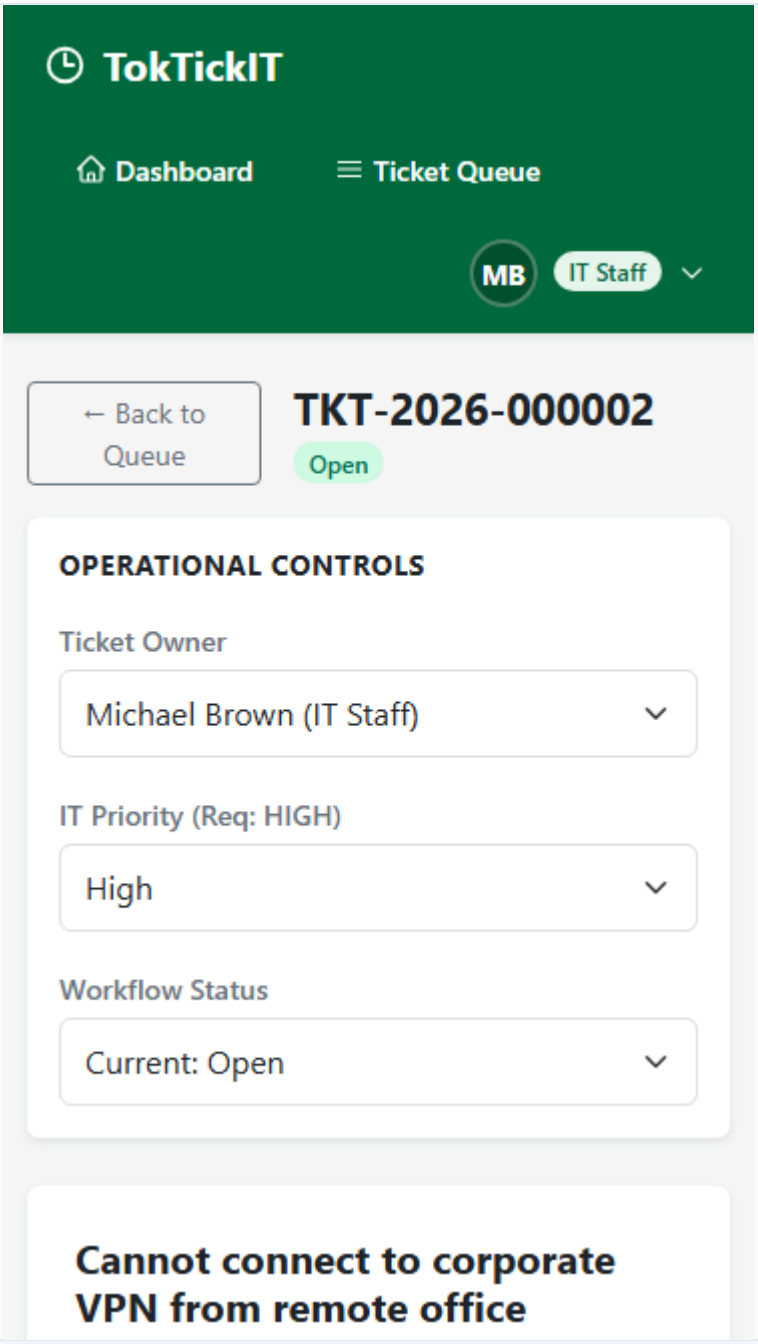



Figure 6.3: Actions Taken section on Mobile viewport (375 × 667) with full touch-target compliance.

## 2. Create/Edit Action Taken Modal

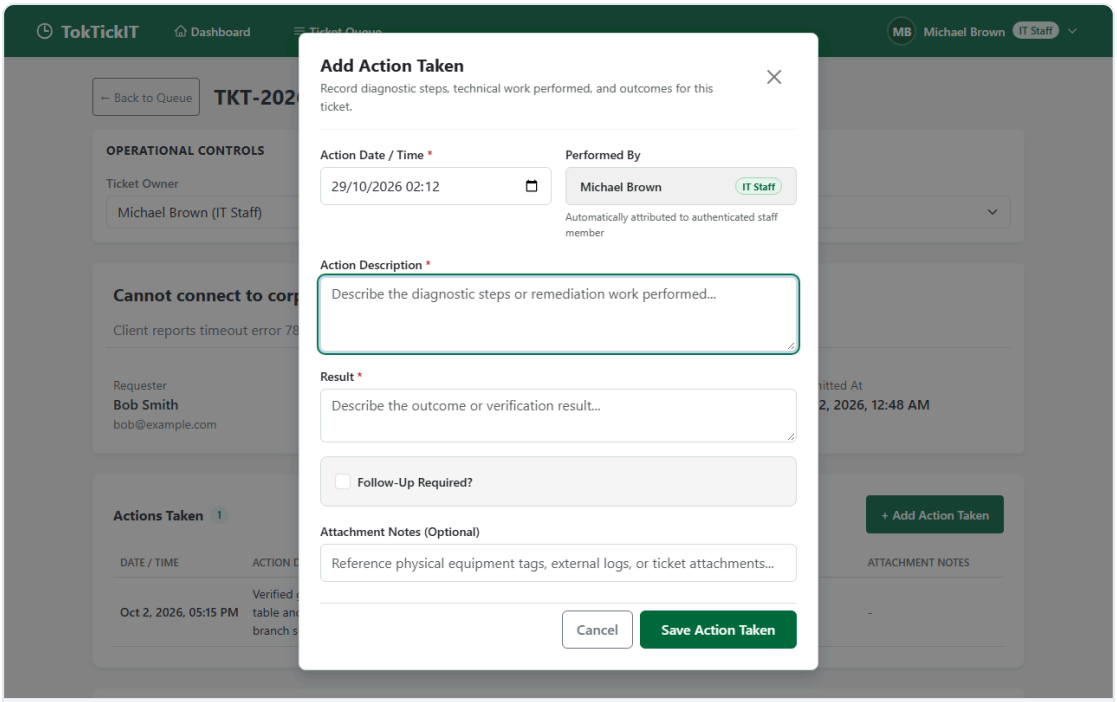



Figure 6.4: Modal overlay with read-only Performer auto-attribution and follow-up validation.

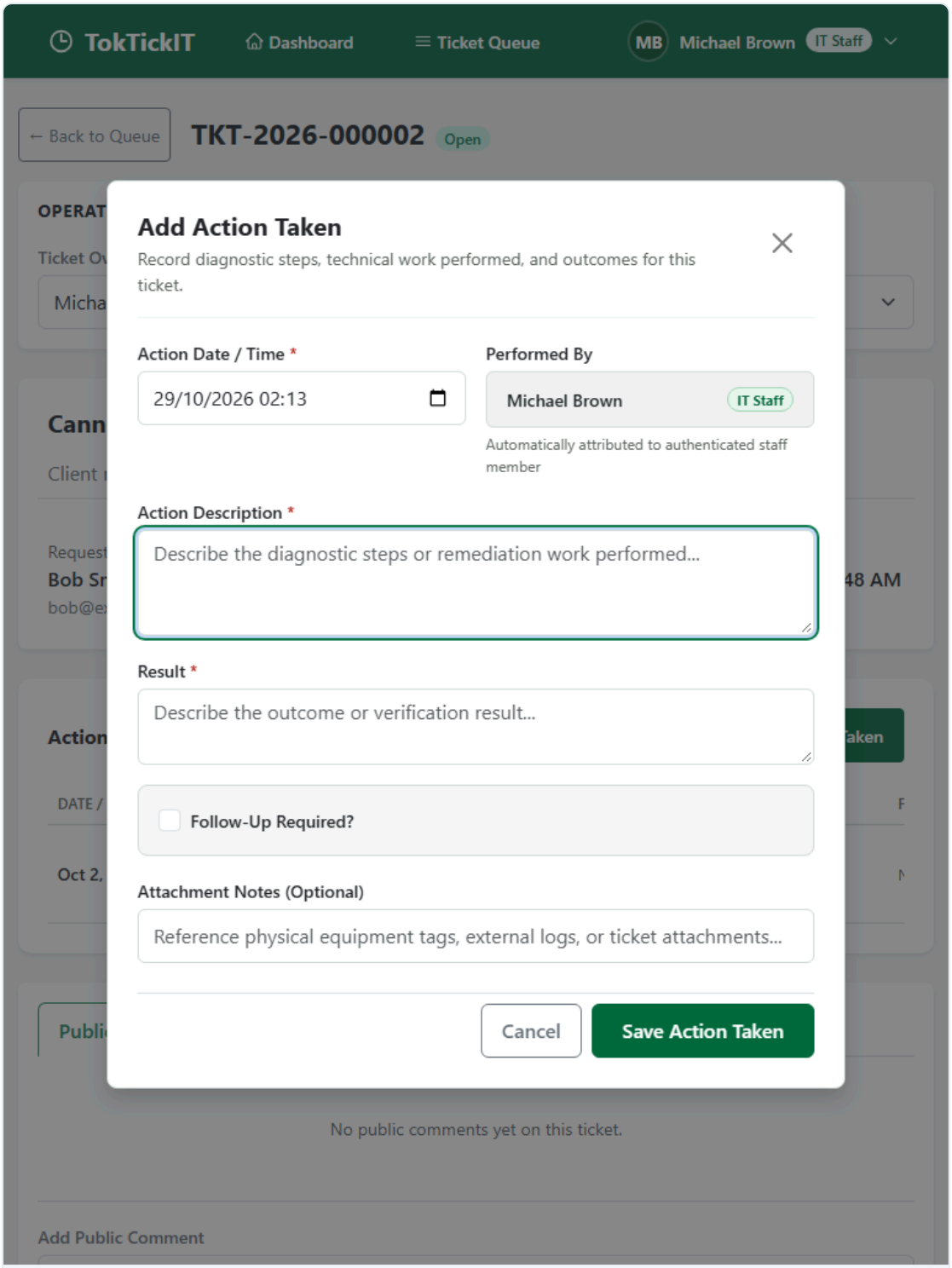



Figure 6.5: Action Taken Modal on Tablet viewport.

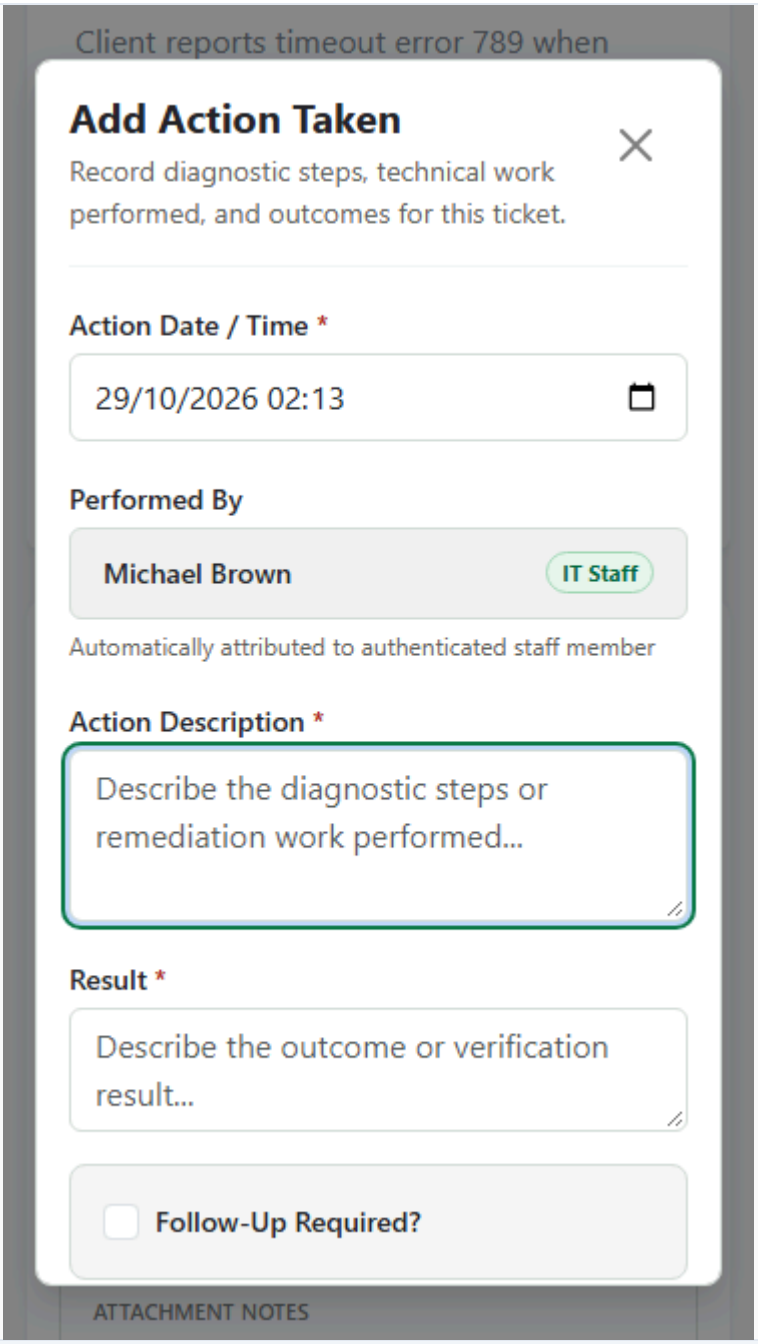



Figure 6.6: Action Taken Modal on Mobile viewport.

### 3. Mandatory Follow-up Validation & Amber Pill

**Follow-up Invariant (BR-05 / VIS-02):** When "Follow-up Required" is checked, a non-empty note is strictly mandatory. Saved actions render a distinct Amber badge ( `#D97706` / `#FEF3C7` ).

### 4. Requester Read-Only Visibility & Terminal Lock

Requesters see Actions Taken in read-only mode without write buttons. Closed and Cancelled tickets lock further action logging.

# Answer Part 7: Working Ticket Workflow

## 1. Resolution Gate & Status Dropdown

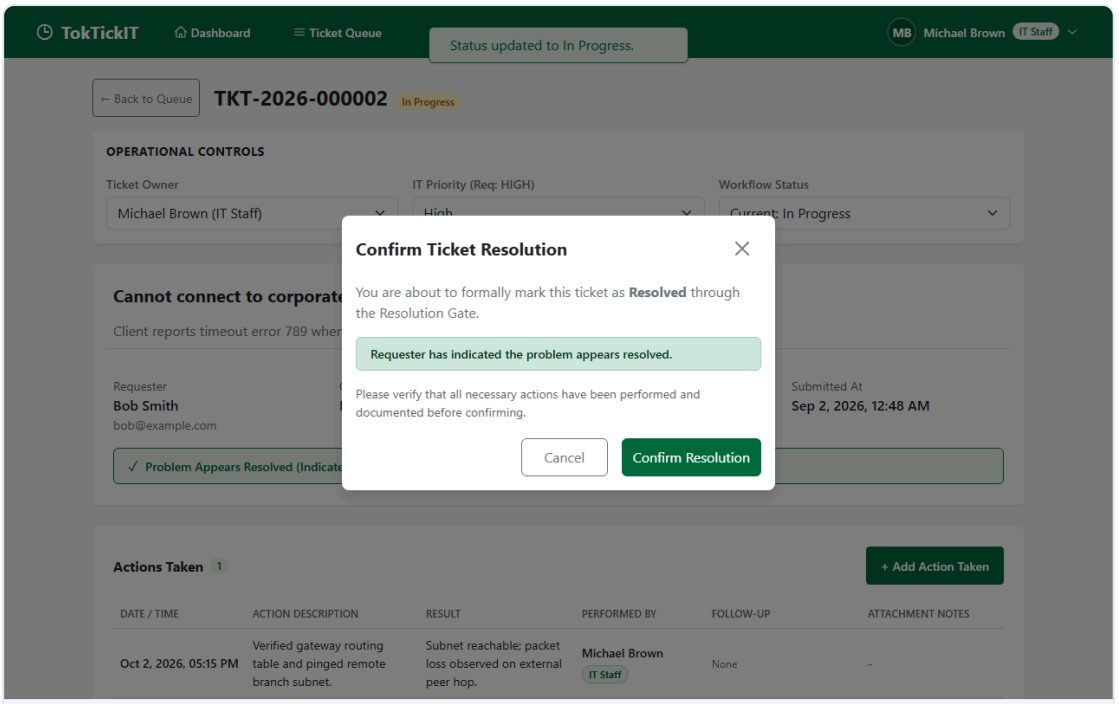



Figure 7.1: Status dropdown restricted to permitted next choices and Resolution Gate confirmation dialog.

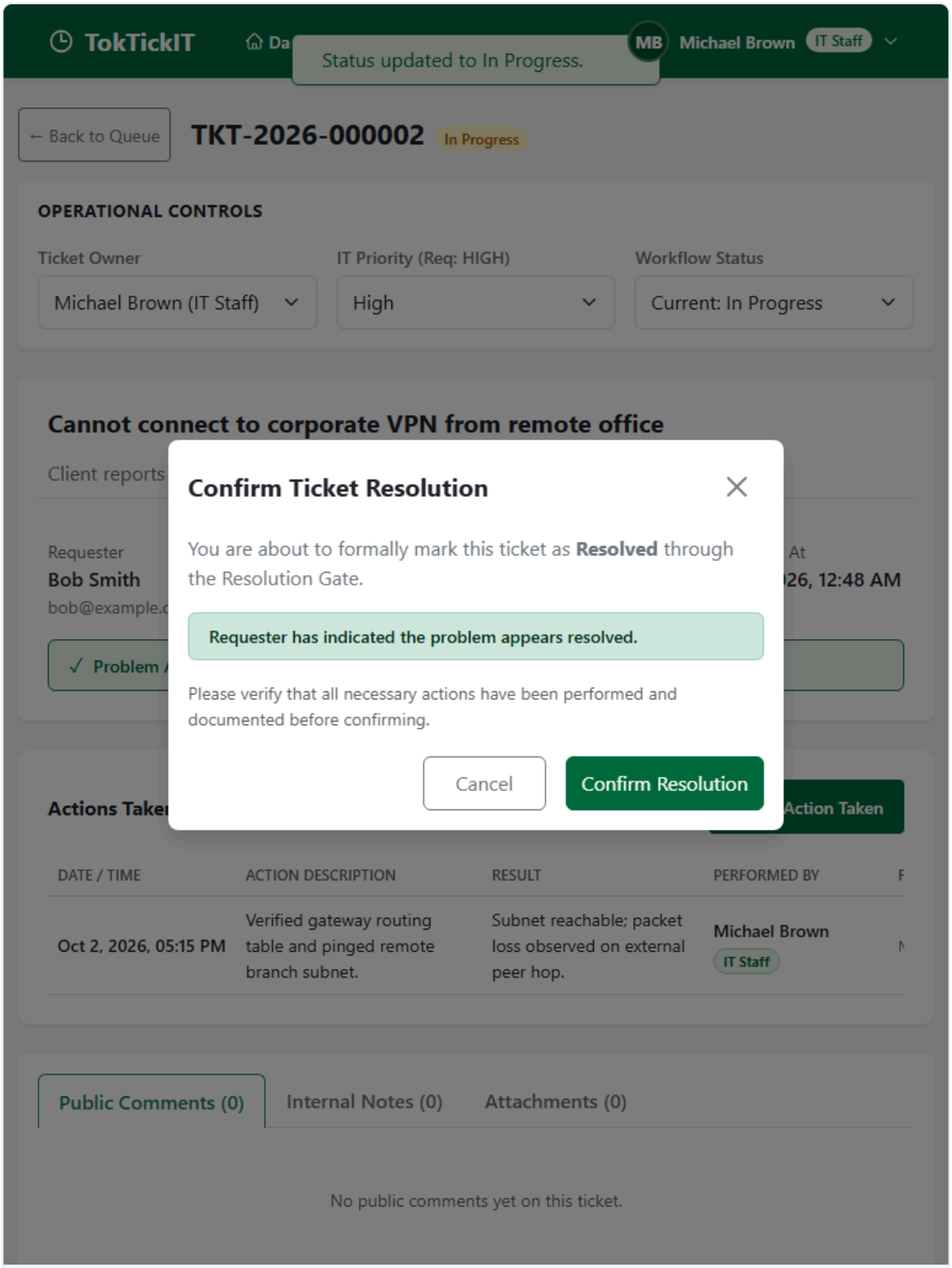



Figure 7.2: Resolution Gate controls on Tablet viewport.

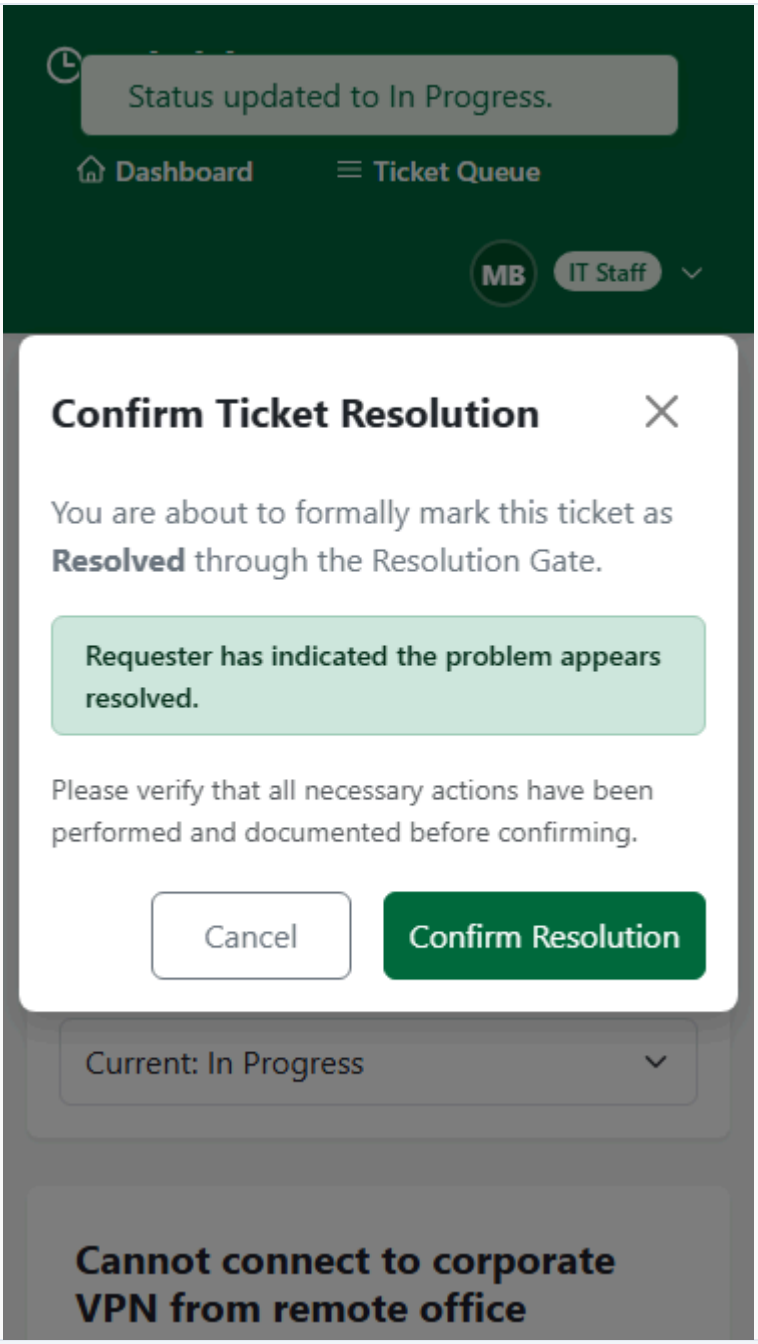



Figure 7.3: Resolution Gate controls on Mobile viewport.

## 2. Requester Advisory Resolution Indicator

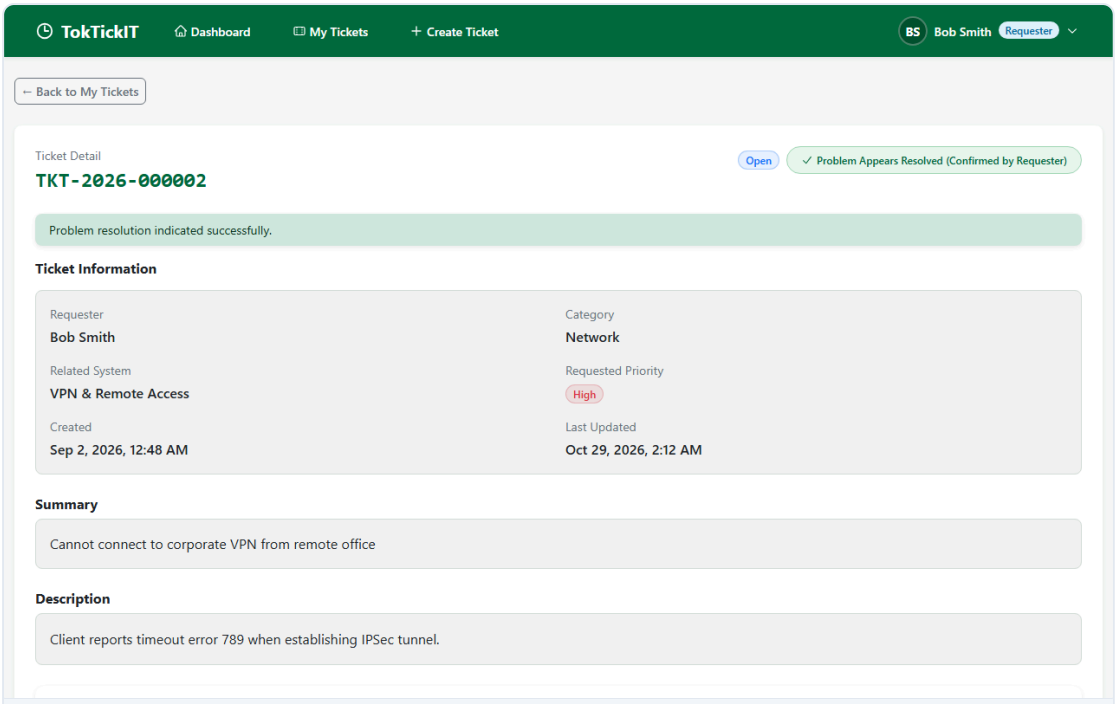



Figure 7.4: Requester indicates "Problem Appears Resolved" — sets advisory flag without altering official status.

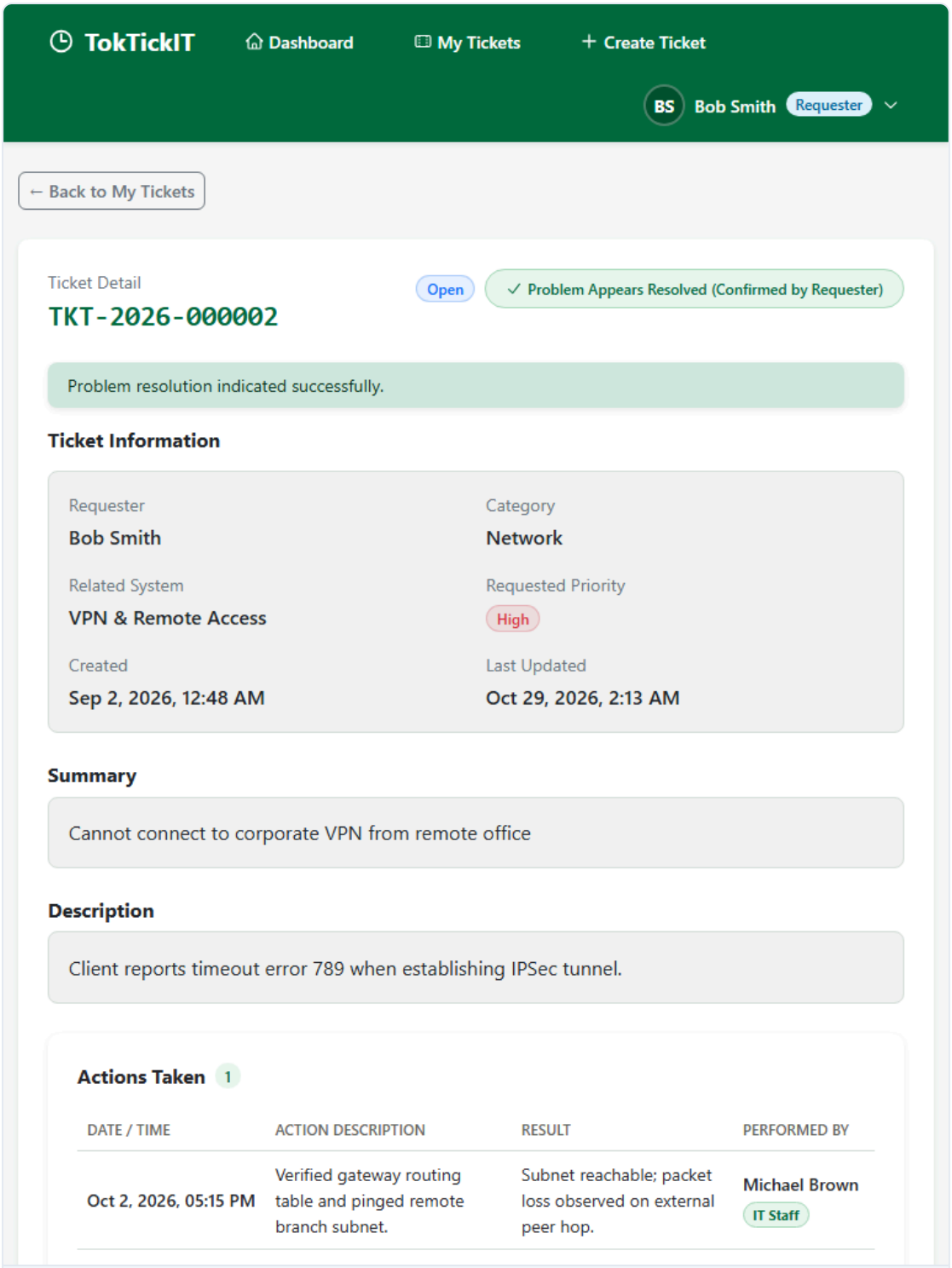



Figure 7.5: Advisory resolution banner on Tablet viewport.

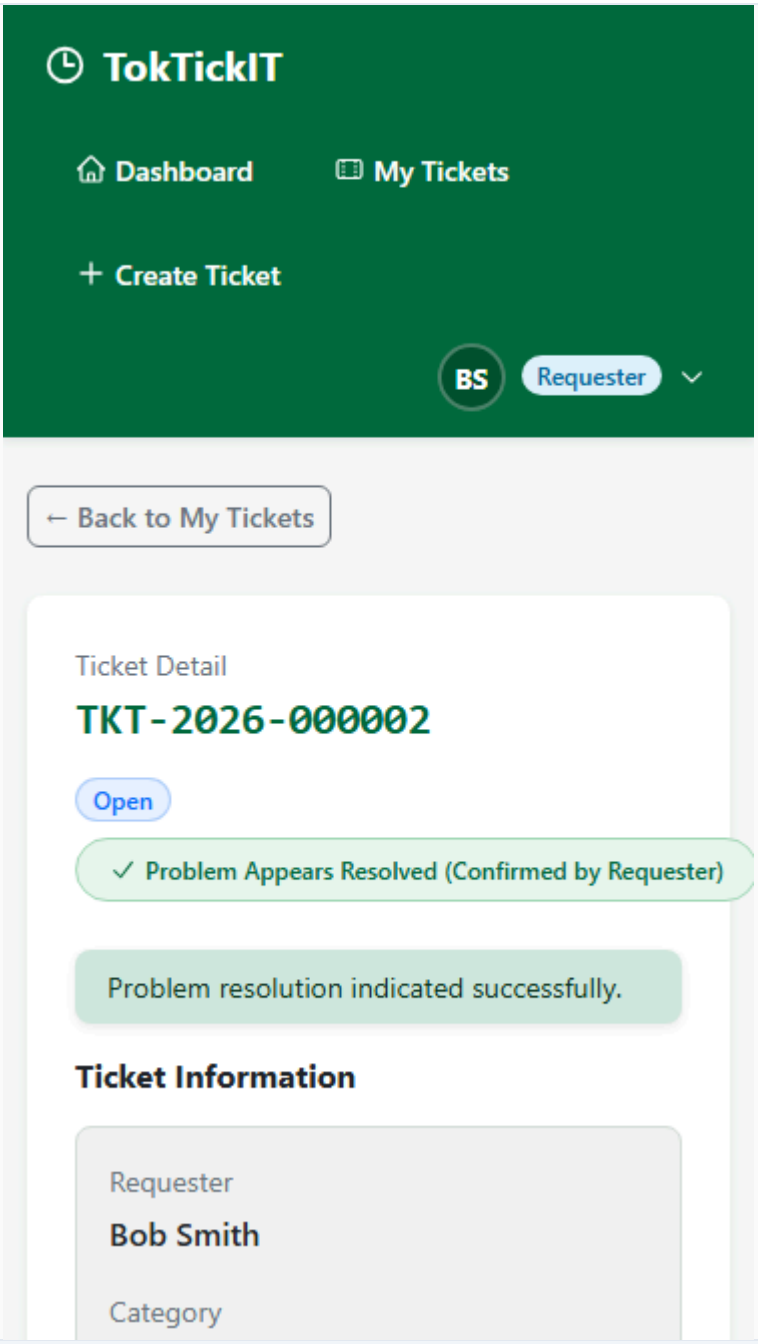



Figure 7.6: Advisory resolution banner on Mobile viewport.

## 3. Optimistic Concurrency Conflict Banner (HTTP 409)

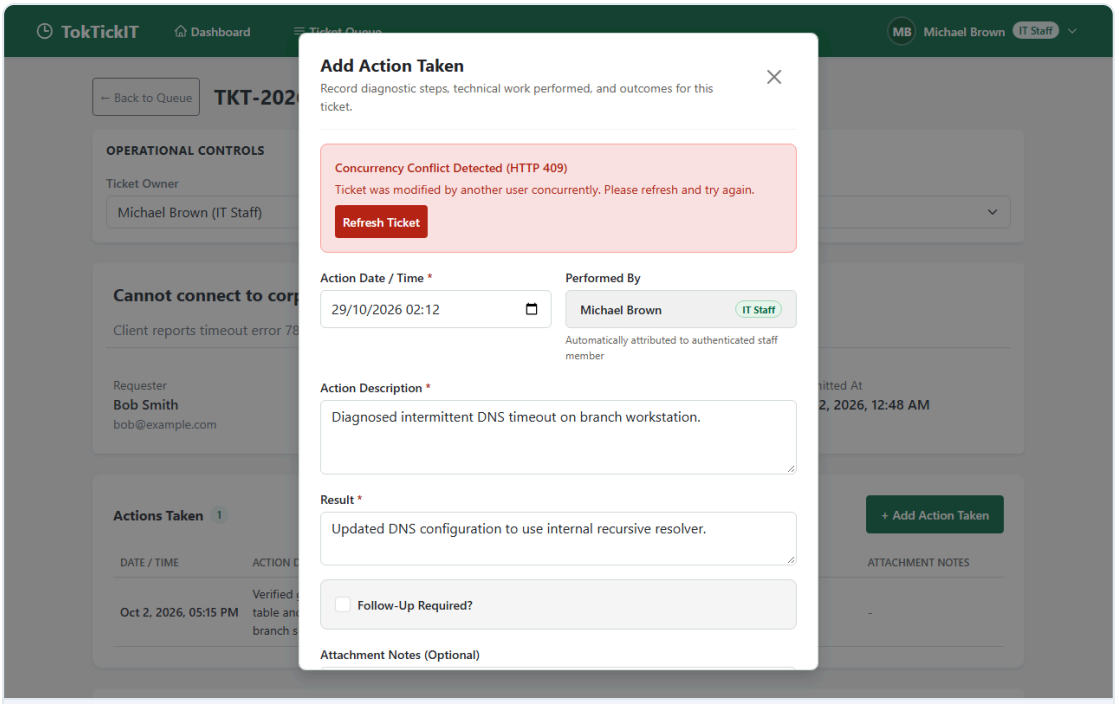



Figure 7.7: Non-destructive conflict banner with "Refresh Ticket" CTA, preserving uncommitted inputs.

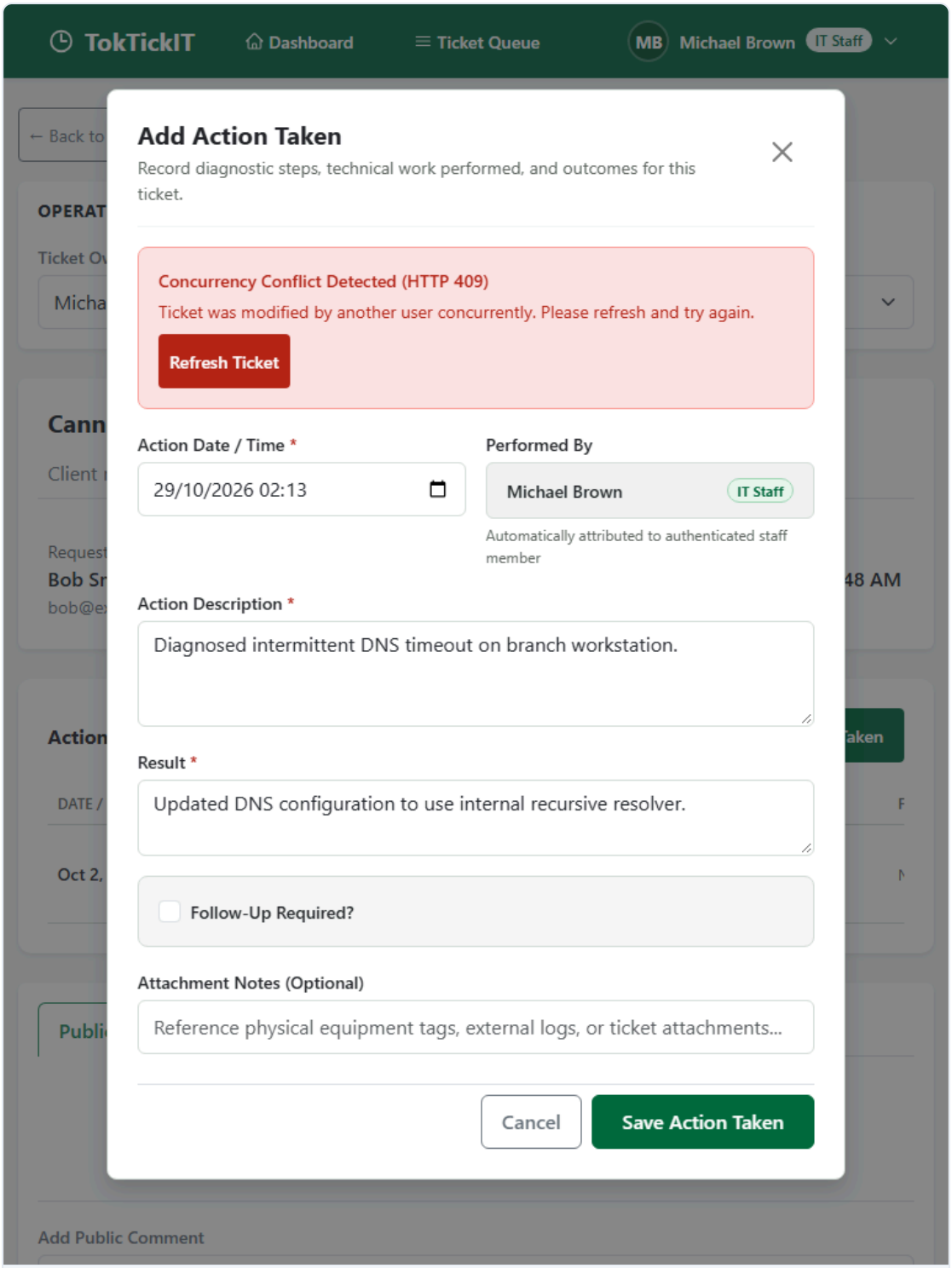



Figure 7.8: Concurrency conflict banner on Tablet viewport.

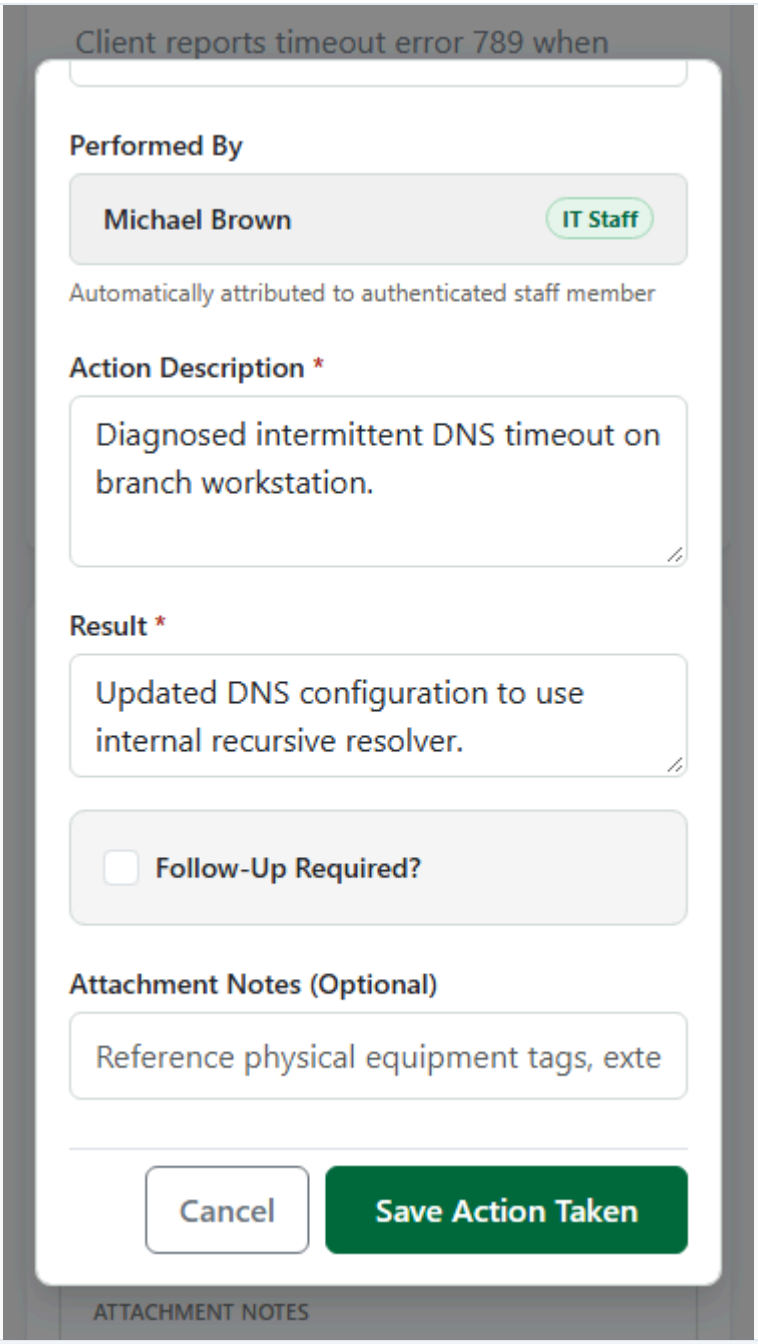



Figure 7.9: Concurrency conflict banner on Mobile viewport.

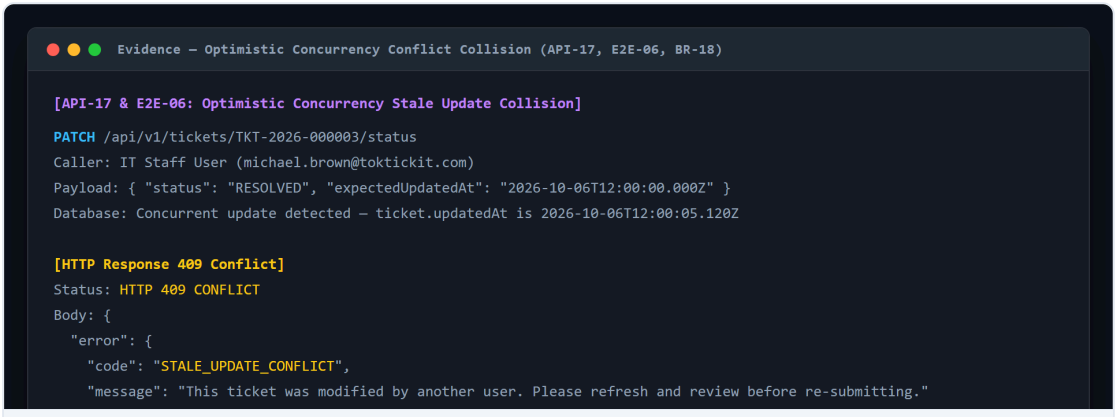



Figure 7.10: Terminal / API evidence of Optimistic Concurrency Conflict Collision (API-17, E2E-06, BR-18).

## Answer Part 8: Working Requester Dashboard & Final Regression UI

### 1. Requester Dashboard Across Viewports

Provides Requesters with personalized metrics: 4 primary cards ( `My Open Tickets` , `Waiting for Requester` , `Resolved` , `Closed` ), recent tickets table, and recently resolved tickets list. All metrics are calculated authoritatively by PostgreSQL and strictly scoped to the logged-in Requester.

| Card Title | Calculation / Included Statuses | Invariant & Relationship Rule | Drill-Down Destination |
|---|---|---|---|
| **My Open Tickets** | `status IN ('NEW', 'OPEN', 'IN_PROGRESS', 'WAITING_FOR_REQUESTER')` | Sum of all active operational tickets owned by the Requester. | `/my-tickets?status=open_all` |
| **Waiting for Requester** | `status = 'WAITING_FOR_REQUESTER'` | **Operational Subset:** Tickets requiring Requester action. Its count is contained within *My Open Tickets*, not a disjoint set. | `/my-tickets?status=WAITING_FOR_REQUESTER` |
| **Resolved** | `status = 'RESOLVED'` | Formally marked resolved by IT Staff. Strictly excludes `CLOSED` . | `/my-tickets?status=RESOLVED` |
| **Closed** | `status = 'CLOSED'` | Terminal tickets. Strictly excludes active `RESOLVED` . | `/my-tickets?status=CLOSED` |

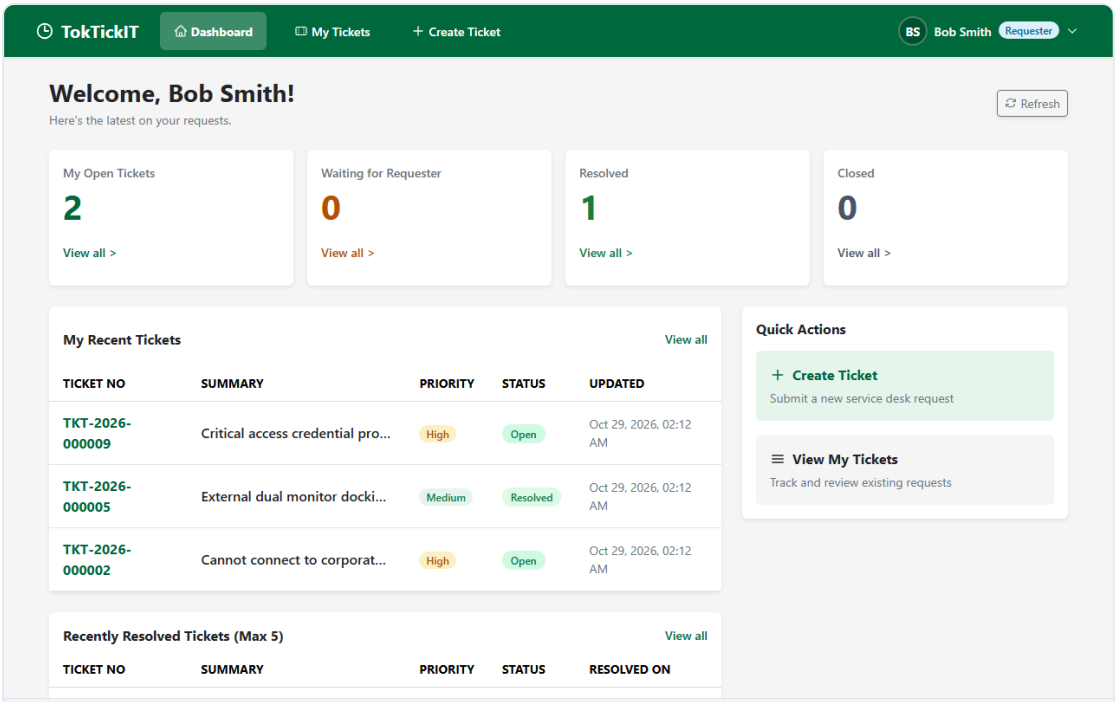



Figure 8.1: Requester Dashboard on Desktop (1280 × 800) with 4 primary metric cards and Recent Tickets table.

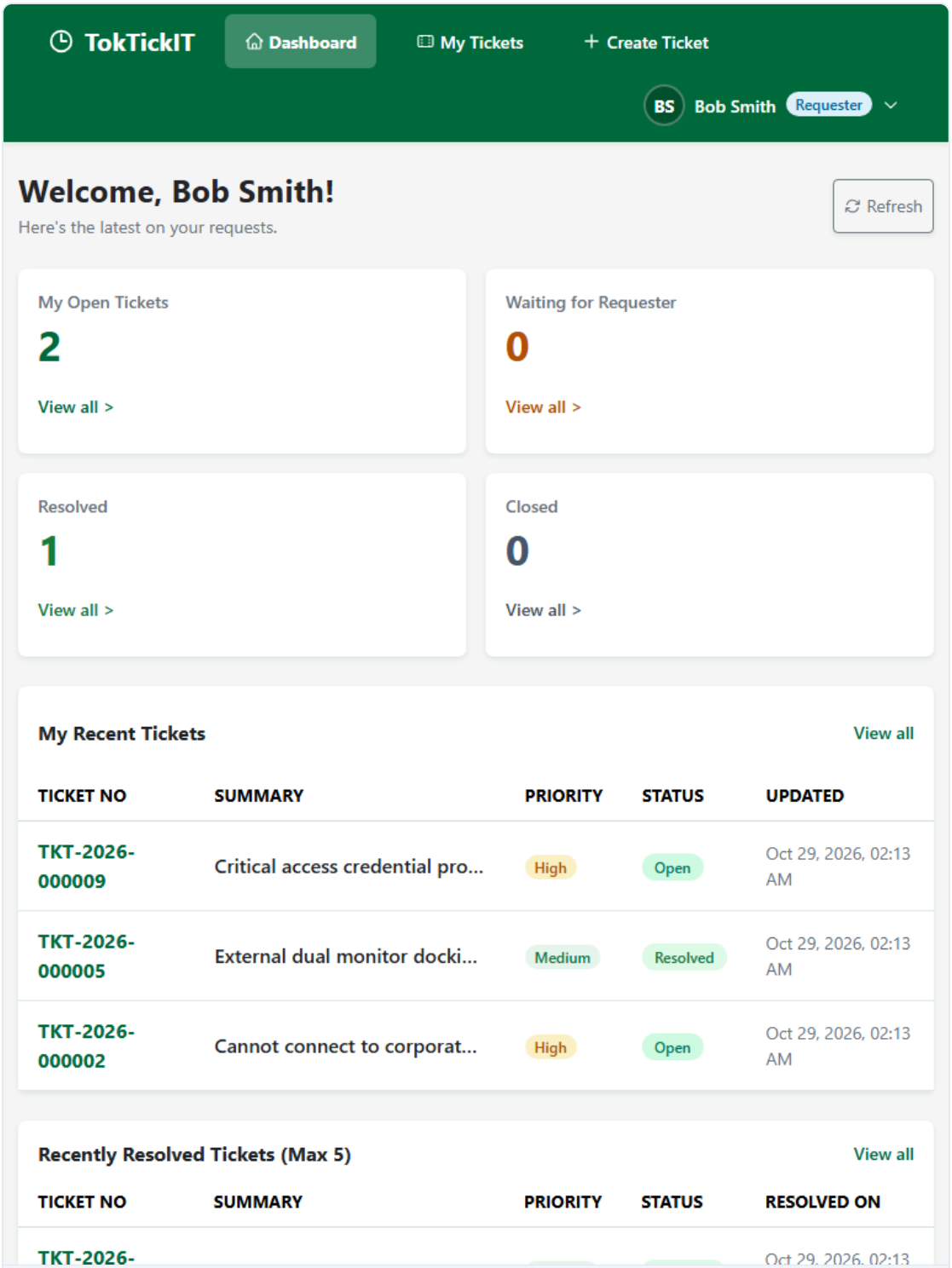



Figure 8.2: Requester Dashboard on Tablet viewport (768 × 1024).

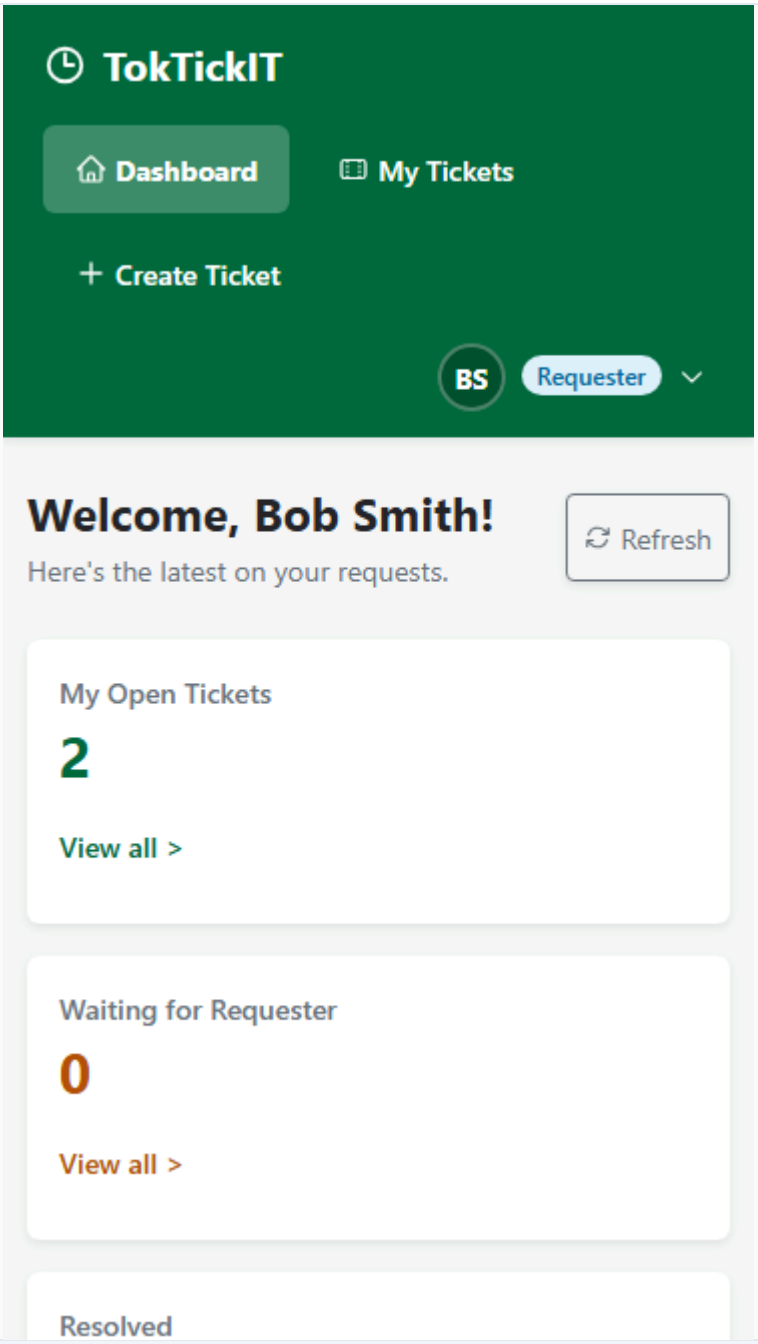



Figure 8.3: Requester Dashboard on Mobile viewport (375 × 667).

## 2. Administrator Dashboard System User Metrics

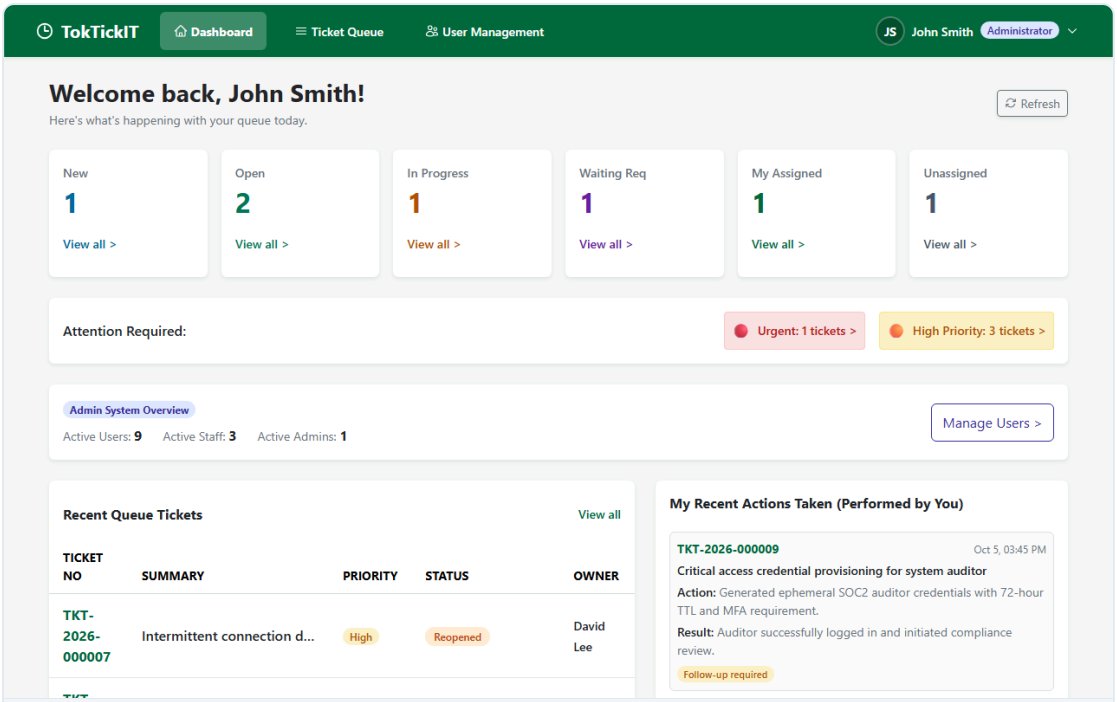



Figure 8.4: Administrator Dashboard displaying staff operational metrics plus System User Accounts summary.

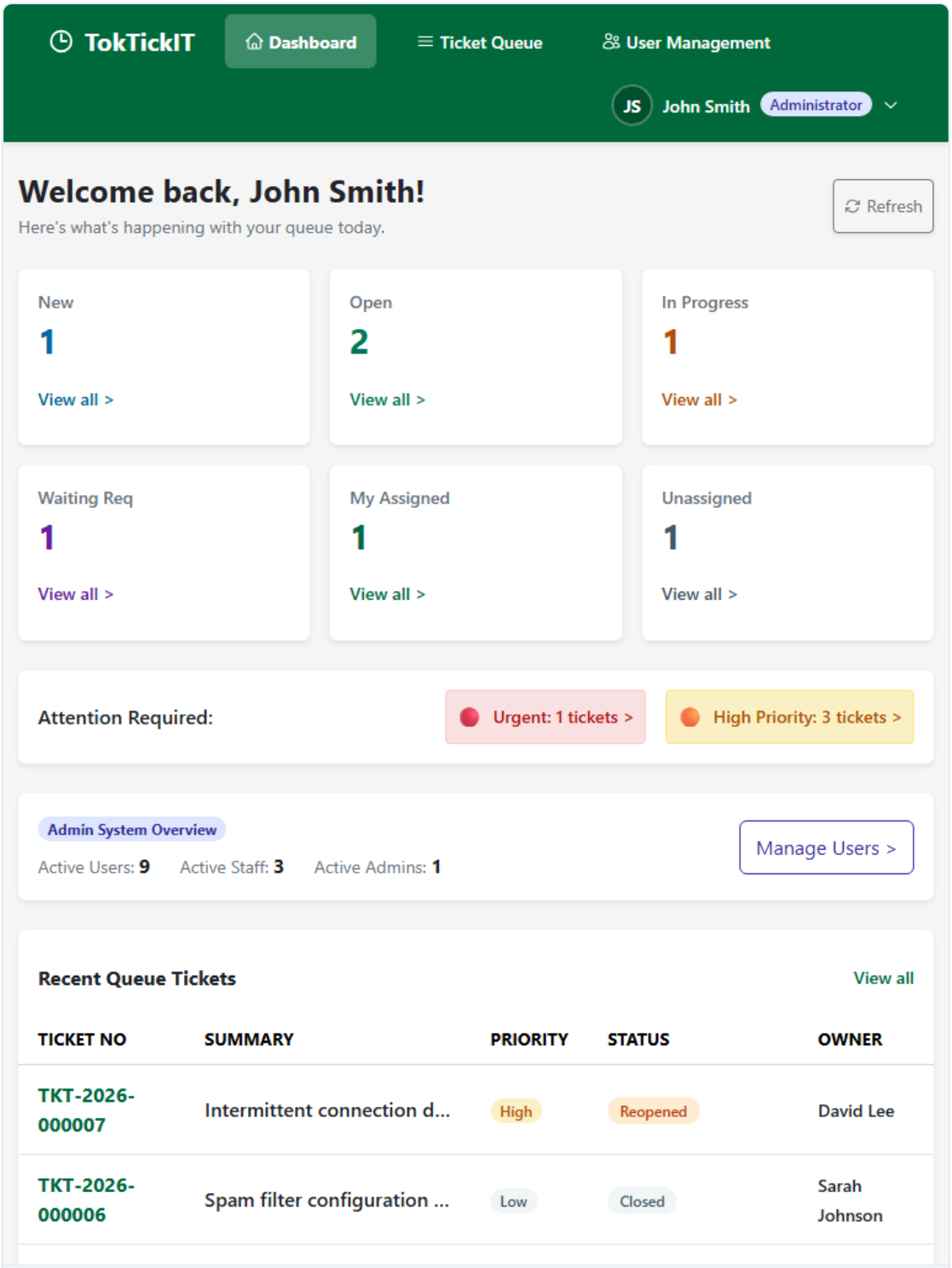



Figure 8.5: Administrator Dashboard on Tablet viewport.

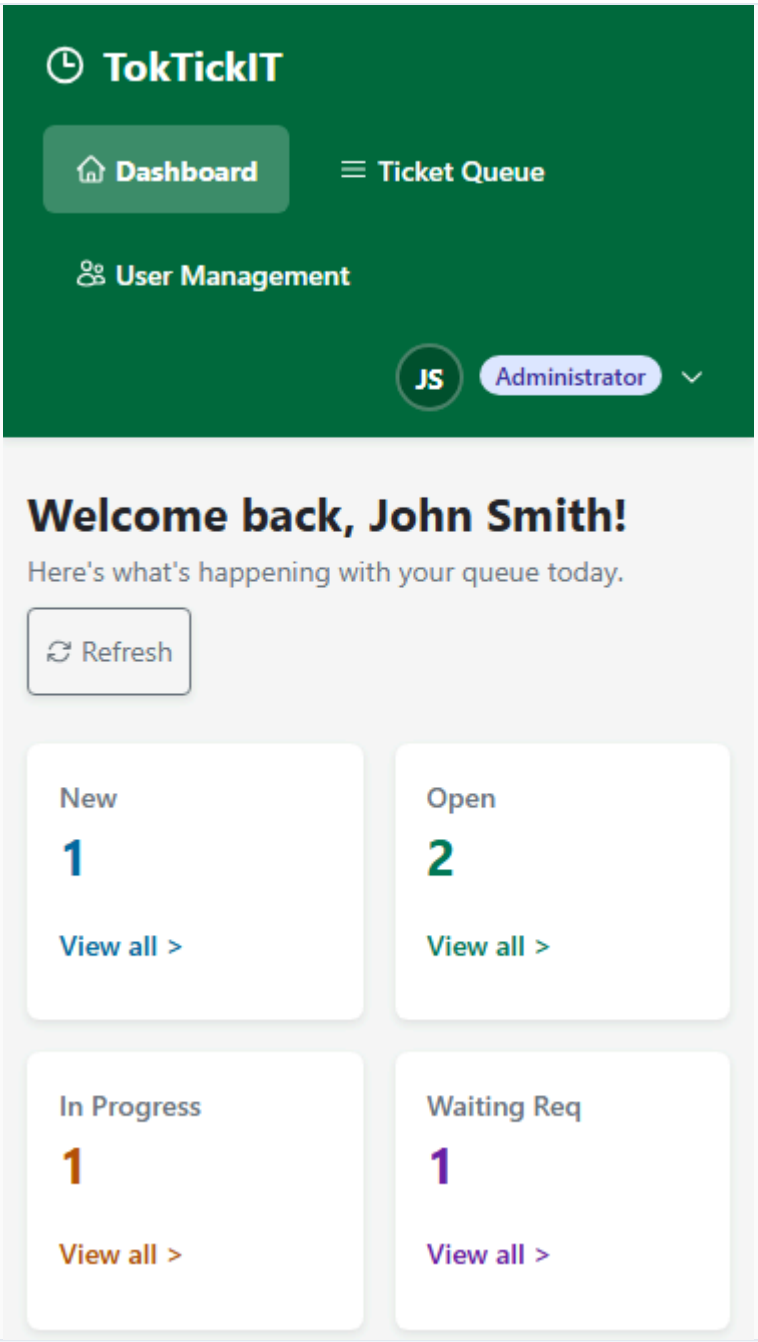



Figure 8.6: Administrator Dashboard on Mobile viewport.

### 3. Full Regression Suite across Labs 1–3 (E2E-10)

Automated test suite `e2e/lab-04/regression.spec.ts` executes 12 multi-viewport regression scenarios covering Lab 1 (health check, categories), Lab 2 (requester creation, My Tickets, detail, attachments), and Lab 3 (auth, forced password change, queue claiming, comments, notes, admin safety) with 100% pass rate.

## Answer Part 9: Zen Green UI, Responsive, Accessibility & Final Polish

**UI Specification Link:** `docs/lab-04/ui-spec.md`

### Design Tokens & Invariant Compliance

- **Zen Green Brand Tokens:** Primary `#006B3C`, Hover `#0B7A46`, Tint `#EAF6EF`, Canvas `#F5F7F6`, White `#FFFFFF`.
- **Amber Follow-up & Notes:** Border `#D97706`, Surface `#FEF3C7`, Text `#92400E`.
- **Editable vs. Readonly:** White inputs (`#FFFFFF`) for active forms; soft gray-green (`#F0F4F2`) for read-only displays.

### Completed Responsive Visual Checklist

| Requirement | Inspection Standard | Verification Method | Compliance Status |
|---|---|---|---|
| **Zero Horizontal Overflow (RESP-04)** | `scrollWidth ≤ innerWidth` across all devices | Automated `assertNoHorizontalOverflow()` | PASS |
| **Minimum Touch Targets (RESP-03)** | Buttons/inputs ≥ 44 × 44px on tablet/mobile | Automated `assertMinimumTouchTargets()` | PASS |
| **Brand Color Tokens (VIS-01)** | Strict adherence to #006B3C Zen Green | Automated CSS token assertions | PASS |
| **Badge Semantic Contrast (VIS-02)** | WCAG AA contrast for Role, Status, and Priority | Automated style tests in `ui-style.test.tsx` | PASS |
| **Visible Focus Rings (VIS-03)** | Visible #0B7A46 focus outline on Tab navigation | Sequential Tab traversal in Playwright & RTL | PASS |
| **Zero Text Clipping (VIS-04)** | Natural word wrapping without truncation | E2E responsive multi-device assertions | PASS |
| **Form State Preservation (BR-18)** | Preserves inputs upon error or 409 conflict | Component tests in `ActionsTakenSection.test.tsx` | PASS |
| **Double-Click Protection (BR-19)** | Disables submit button and shows spinner in-flight | E2E and component race condition tests | PASS |

★ Visual QA & Responsive Invariants Checklist (VIS-01..04, RESP-01..04, BR-18..19)

TokTickIT Sprint 4 Verified

| Invariant Code | Visual & Responsive Requirement | Verification Method | Result |
|---|---|---|---|
| **RESP-01** | Desktop layout at 1280×800 with full tables, cards, and multi-column navigation | Playwright Desktop viewport + Visual QA | PASS |
| **RESP-02** | Tablet layout at 768×1024 with 2-column responsive grid and modal auto-centering | Playwright Tablet viewport (AC-23) | PASS |
| **RESP-03** | Mobile layout at 375×667 with stacked cards and touch targets ≥ 44 × 44px | assertMinimumTouchTargets(page) | PASS |
| **RESP-04** | Zero horizontal overflow across all viewports (scrollWidth ≤ innerWidth) | assertNoHorizontalOverflow(page) | PASS |
| **VIS-01** | Zen Green brand token adherence (#006B3C, #0B7A46, #EAF6EF, #FFFFFF) | Computed CSS token assertions in RTL & Vitest | PASS |
| **VIS-02** | Distinct Amber theme for Follow-up Required (#D97706, #FEF3C7) and Internal Notes | Automated style assertions in ui-style.test.tsx | PASS |
| **VIS-03** | High-contrast visible focus outline (#0B7A46, 2px) on sequential Tab key traversal | Sequential Tab key traversal in E2E & RTL | PASS |
| **VIS-04** | Zero text clipping, word wrapping on long text, and responsive metric cards | E2E responsive multi-device assertions | PASS |

Figure 9.1: Visual QA and responsive design invariant verification matrix.

## Complete 24 Screenshot Evidence Inventory

All 24 screenshot evidence artifacts captured across Desktop ($1280 \times 800$), Tablet ($768 \times 1024$), and Mobile ($375 \times 667$) are committed and verified under `artifacts/lab-04/screenshots/`.